# Manipulating Disclosure Tone: Understanding Acquiring Firms' Strategies in Stock-for-Stock Mergers and Acquisitions

*Short title*
*Disclosure tone management around stock-for-stock M&As*


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## ABSTRACT

Based on a sample of 5,558 stock-for-stock mergers and acquisitions (M&As) between 2004 and 2020, we show that, in the year before the M&A announcement, stock bidders manipulate the narrative content of corporate disclosures by inflating the tone of earnings press releases in order to curtail the target acquisition cost. Furthermore, tone inflation is more pronounced in firms executing larger-scale deals, firms with weaker historical performance, and firms with limited monitoring mechanisms in place. We also find that, although tone inflation propels an upswing in the bidder's stock prices and leads to a reduction in the overall target acquisition cost, it exposes the bidder to unfavorable long-term returns after the merger and acquisition announcement. We subject our analyses to an exhaustive array of robustness and endogeneity tests, which consistently affirm the validity of our conclusions. By distorting the narrative tone, stock bidders compromise the quality and reliability of information provided to stakeholders, undermining the transparency and trust upon which effective financial reporting relies.


**Highlights**

1. Stock-for-stock mergers and acquisitions (M&As) strategically inflate the tone of earnings press releases during the year preceding the M&A announcement, aiming to diminish the target acquisition cost.
2. The analysis reveals a significant surge of 15.32% in the tone of earnings press releases during the pre-merger period, resulting in substantial estimated savings of $21.777 million for the average stock bidder in acquisition costs.
3. The degree of tone manipulation exhibits variations based on specific factors: larger deals, firms with lower historical performance, and firms lacking robust monitoring mechanisms experience a stronger inclination towards tone inflation.
4. A consequential outcome of this deliberate tone manipulation is the emergence of negative long-term returns subsequent to the M&A announcement.
5. We also find that, prior to the announcement of a merger or acquisition (M&A), bidders tend to manipulate the corporate narrative structure by deliberately amplifying positive tone dispersion within their earnings press releases, thereby enhancing the overall impact of their tone inflation strategy.


**JEL Classification :** M13, G32, N20, O57, L26

**Keywords:** M&As, Disclosure tone management, Earnings press releases, Textual analysis



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**Availability: Data is available through the public sources cited in this paper. The earnings press releases are available on the SEC EDGAR platform, while the accounting and finance variables were retrieved from the WRDS platform. Mergers and acquisition data were obtained from the Securities Data Company's (SDC's) Mergers and Acquisitions database.**

**Declaration of Interest:** None

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## 1. Introduction

Although mergers and acquisitions (M&As) activity has significantly grown over the last decade, there remains much about the M&A process that we need to comprehend, including the implication of the method of payment on managers' opportunistic behavior.[1] The use of stock as a method of payment is also known to create a strong incentive for managers to manipulate earnings to positively influence the stock price, and therefore reduce the target's acquisition cost (Erickson and Wang, 1999; Louis, 2004). However, it remains unclear whether stock bidders also strategically vary the narrative content of their corporate disclosures. Given the substantial influence of qualitative information on the firm's stock price (Henry, 2008; Henry and Leone, 2016; Fu, Wu, and Zhang, 2021; Arslan-Ayaydin, Boudt, and Thewissen, 2016), it is essential to examine whether the use of stock as a form of payment leads managers to manipulate the content of qualitative information in corporate disclosures.

In the majority of M&As, the acquiring firm buys the target's assets using cash. However, in some cases, the acquiring firm uses its stock in exchange for the target's stock. In stock-for-stock mergers, acquiring firms exchange their stock against the stock of the target at a predefined rate. This means that the higher the price of the acquiring firm's stock on the agreement date, the fewer the number of shares that must be issued to purchase the target firm. Prior literature shows that such a negative relationship between bidders' stock price and shares issued in the transaction leads bidders to increase their own premerger stock price to lower the target's acquisition cost. For instance, Erickson and Wang (1999), Louis (2004) and Higgins (2013) find that acquiring firms tend to manipulate total accruals upwards in the periods prior to the merger announcement, while He, Liu, Netter, and Shu (2020) find that bidders significantly manage down analyst earnings forecasts prior to merger announcements. Overall, prior research provides strong evidence that the information environment around stock-for-stock mergers is characterized for being highly uncertain and subject to opportunistic manipulation.

[1] The method of payment in a M&A transaction can either be cash, stock (in which an exchange ratio is specified for conversion of target shares into acquirer shares), or a combination of the two (mixed deals).

Yet, there remains limited research on whether bidders in stock-for-stock mergers tend to manipulate the content of corporate disclosures to increase their stock price. Disclosure tone measures both what and how information is disclosed (Loughran and McDonald, 2011), reflecting both firms' performance and managers' disclosure incentives (Fu et al., 2021). A growing body of literature highlights that the narrative section of corporate disclosures provides incremental information to financial numbers and management forecasts in explaining the firm's future cash flows (Henry, 2008; Henry and Leone, 2016; Baginski, Demers, Kausar, and Yu, 2018; Arslan-Ayaydin et al., 2016; Cho, Phillips, Hageman,and Patten, 2009; Cho, Roberts, and Patten, 2010; Davis, Piger, and Sedor, 2012). However, managers can also use the tone as a tool to mislead stakeholders. Arslan-Ayaydin et al. (2016) show that CEOs' equity-based incentives leads managers to inflate the tone of earnings press releases, while Henry, Thewissen, and Torsin (2021) find that a cautious disclosure tone mitigates potential home bias-related credibility and asymmetric information concerns arising from cultural and institutional distance. In this paper, we therefore extent prior research and propose that, in order to increase their stock price and reduce the acquisition cost of the target, bidders in stock-for-stock mergers inflate the tone of corporate disclosures.

It is not *ex ante* obvious whether the potential gains outweigh the costs of inflating the tone of their corporate disclosures. First, it is unclear whether acquirers would be willing to exaggerate the use of positive news at the cost of reducing their credibility with regard to investors. That is, if the manager's optimism is inappropriate, investors may be potentially misled in terms of the current and future earnings and firm value of the newly combined corporation. The company's credibility may be jeopardized (Yuthas, Rogers, and Dillard, 2002), along with the reliability, value, and trust in their corporate communication. In addition, litigation risks are a significant cost associated with tone inflation in qualitative disclosures. According to Rogers, Van Buskirk, and Zechman (2011a), the use of optimistic language by managers heightens litigation risk. Their research indicates that plaintiffs often target optimistic statements in their lawsuits and that companies facing lawsuits tend to have unusually optimistic language in their earnings announcements. Therefore, the fear of litigation due to overly optimistic voluntary disclosures can discourage managers from being overly optimistic in their communications, especially during high-profile events like stock-for-stock mergers. In fact, Ball and Shivakumar (2008) question the notion that managers manipulate earnings upwards before stock issuance because of the high market scrutiny and litigation risk that characterize these events. Finally, the target and its advisors are informed professionals and users of accounting information. As such, they are likely to be familiar with the tone inflation used by the bidder. In light of this manipulation of corporate narratives, the target may oppose and reject the deal.

Against this backdrop, whether stock-for-stock acquirers engage in tone manipulation is not only an empirical question, but also raises ethical concerns as effective information disclosure and information transparency are key aspects of corporate ethical behavior that could build stakeholder trust and sustainable competitive advantage (das Neves and Vaccaro, 2013). Despite being exposed to several costs associated with the manipulation of narrative disclosures, the question is therefore whether stock-for-stock bidders engage in the upward manipulation of the tone of their voluntary corporate disclosure. This question is of interest, given that both quantitative and qualitative features are used to evaluate managerial abilities and firm performance (Amir and Lev, 1996; Arslan-Ayaydin, Bishara, Thewissen, and Torsin, 2020). We start by examining the use of tone management before M&A announcements, and specifically, whether bidders in stock mergers are more likely to inflate the tone of earnings press releases. We then conduct additional cross-sectional tests to analyze the impact of the acquiror's pre-merger performance, deal value and firm monitoring on bidders' propensity to manipulate qualitative information. We further study how this manipulation of qualitative content,if used, impacts bidders' stock prices and, in turn, their acquisition costs.

We use a sample of 5,558 M&A deals between 2004 and 2020, including 302 stock deals, 4,088 hybrid deals (takeovers that use both cash and stock) and 1,168 cash deals. To measure the linguistic tone, we follow prior literature (see, e.g., Henry and Leone, 2016) and compute the spread in the number of positive and negative terms in the earnings press release from the Henry (2008) lists of words. We follow prior M&A literature and examine quarterly earnings press releases of bidder firms over five quarters before the merger announcement. We find that the tone of earnings press releases for stock-for-stock mergers significantly increases in the four quarters before the M&A announcement quarter to reach a peak two quarters ahead of the M&A announcement. In fact, the median (average) tone increases by 15.320% (4.447%) between quarter -6 and quarter -2. In the quarter of the M&A announcement and the subsequent quarters, we observe that the tone of earnings press releases returns to its original level and no longer significantly differs from that of cash-based deals.

In contrast, we find that the tone of earnings press releases of cash deals does not significantly change over the five quarters before the M&A announcement, which is in line with the lack of incentive to engage in tone management in these deals. Controlling for a series of deal and firm characteristics, as well as year and industry fixed effects, the multivariate analysis confirms that the tone of stock deals significantly increases before the M&A announcement, relative to cash deals. Our analyses further indicate that the degree of tone inflation increases with the deal size, but decreases with the presence of control mechanisms in the firm and the firm's past performance.

Overall, our results are consistent with the strategic manipulation of tone by bidding firms in the period prior to the announcement of the merger agreement. To further ensure that our results are not driven by confounding factors, we conduct a number of identification tests for alternative explanations. We first examine the impact of earnings manipulation and explore whether the upward effect of stock deals on tone is a separate effect and cannot be explained by upward earnings management (Erickson and Wang, 1999). If the more optimistic tone of firms is mostly driven by the manipulation of earnings numbers, the impact of stock deals on the content of earnings press releases should become statistically insignificant once we control for earnings management. Contrary to this perspective, our findings reveal that the relationship between stock deals and tone continues even with upward earnings management. This indicates that stock deals cause managers to consistently manipulate the content of earnings press releases, independent of their earnings management practices.

A second potential concern is that unobserved characteristics correlated with the method of payment explains the tone of earnings press releases. We therefore correct for this potential endogeneity between tone and the method of payment by conducting five tests to exclude the possibility that alternative variables explain the impact of the method of payment on the tone. We first follow Fisch and Momtaz (2020) and employ a restricted control function regression. We then exploit an instrumental variable approach using investors' ownership as instrument. We then also compare stock and cash bidders using the propensity score matching (PSM) and entropy-balancing methods. Finally, we conduct two pseudo-events studies by shifting merger announcements either forward or backwardby four quarters and repeat our baseline regressions. Although no approach can perfectly rule out endogeneous relationships, our results from our endogeneity tests provide supporting evidence that bidders' tone inflation is likely to be caused by the use of stock payment and not by omitted bidder-, target-, deal-specific characteristics.

Based on this causal evidence, we next estimate the impact of inflating the tone of earnings press releases on stock bidders before the merger. To do this, we conduct several regressions of a bidder's abnormal stock return around earnings announcements on the abnormal tone in earnings press releases. Controlling for other firm-, year- and industry-specific characteristics, we find that increasing the tone of earnings press releases is significantly associated with an increase in pre-merger stock prices. In fact, we estimate that tone inflation leads to savings in acquisition costs of $21.777 million for an average stock bidder in the pre-merger period. In addition, we examine the impact of this tone inflation on bidders' abnormal stock returns around and after the merger announcement. If the tone inflation is opportunistic, we expect the abnormal tone in earnings press releases to cause a temporary overpricing of bidders' stocks at the M&A announcement and its effect to reverse in the long run. In line with this prediction, our results show that stock-for-stock bidders who engage in tone inflation benefit from superior returns at the M&A announcement. However, in the long-term, we document that stock bidders suffer from worse long-term returns. This evidence is in line with prior research on earnings management, which documents that firms that manipulate earnings to beat expectations are still able to benefit from higher short term returns, but suffer from long term losses (Bartov, Givoly, and Hayn, 2002; Sloan, 1996).

Our study adds to the literature in several ways. First, motivated by prior research on the relation between disclosure tone and stock valuation, our study provides novel empirical evidence on the impact of M&As' method of payment on managers' strategic motives. In particular, we contribute to Erickson and Wang (1999), Higgins (2013), He et al. (2020) and Louis (2004) by showing that influencing investors' decisions is not only achieved by manipulating hard financial information, but also the content of disclosure narratives. As such, the management of tone should be regarded as an independent and distinct manipulation tool utilized by acquirers in an attempt to inflate the stock price around stock-for-stock M&As.

Second, this paper adds to prior literature on voluntary disclosure by examining how important economic events influence the quality of corporate narratives (see, e.g., Erickson and Wang, 1999; Louis, 2004; He et al., 2020; Henry et al., 2021). Instead of focusing on the identification of new methods to optimize the definition of tone in corporate disclosures (Jegadeesh and Wu, 2013; Loughran and McDonald, 2011; Boudt and Thewissen, 2019), our study aims at highlighting the strategic motives that may lead managers to manipulate the content of such information and how the stock market responds to such manipulation. Our focus on one of the most important events in a company's lifetime (e.g., M&As) clearly illustrates that the content of corporate narratives has both information-sharing purposes and strategic motives to influence the stock price (Yan, Aerts, and Thewissen, 2019). As such, our results help investors understand the informational dynamics around stock-for-stock M&As and better comprehend the influence of managerial incentives on corporate reporting and the firm's value in the short and long term around M&A events.

Third, our paper highlights an important ethical concern with respect to the quality of the narratives contained in corporate disclosures. Our findings indicate that acquirers put forward events that serve at inflating the stock price, rather than being accountable to the firm's stakeholders and to the society. As such, the opportunistic use of corporate disclosures as an instrument to construct reality in a way that benefits bidders at the expense of the target represents an ethical issue, as "it sustains relations of domination" (Beelitz and Merkl-Davies, 2012, p. 102). Many questions have been raised concerning the ethical propriety of these relations of domination in the context of M&As. Most of these relate to the tactics used by the target when trying to avoid being acquired (for a discussion, see; Werhane, 1988), such as the poison-pill tactics, the institution of golden parachutes, the granting of lock-up options and the anti-trust challenges. Instead of focusing on the tactics used by the target firm, our paper focuses on unethical methods used by bidders during stock-for-stock M&As to reduce the acquisition price. The main ethical implication of such tone inflation is that, if discretionary narrative disclosures are used to strategically inflate the stock price rather than incremental information purposes, then financial reporting quality will be undermined. Given that we show that investors are susceptible to it, then adverse capital misallocations may result. Such manipulation of corporate disclosures' qualitative information is therefore an important corporate governance and regulatory issue. Through this paper, our findings address this issue by understanding the incentives that influence managers in manipulating their disclosures. In particular, we extent the findings of Erickson et al. (2008), who discover that managers manipulate earnings figures to present a favorable image of their performance prior to stock M&As. We demonstrate that, irrespective of managers' tendency to manipulate earnings numbers, they strategically craft the content of the narrative sections of corporate disclosures before the M&A announcement.

## 2. Literature review and hypothesis development

### *2.1.* Stock-for-stock M&As and earnings management

The total number of shares issued by the acquiring firm is determined by an exchange ratio, which is the number of shares of the acquiring firm's stock to be issued for each share of the target's stock, as negotiated by the acquirer and the target. For example, the exchange ratio is four if the acquirer agrees to exchange four of its stocks for one target firm share. This ratio is defined such that target shareholders generally obtain a significant premium above the current market price (Erickson and Wang, 1999). In order to get to the exchange ratio, the acquiring firm usually first defines the offer to the target firm by specifying a fixed dollar value amount for each target share. The number of stocks of the acquiring firm's shares exchanged for each of the target's shares is then calculated based onthe acquiring firm's stock price near the merger agreement date.[2] As a result, there exists a negative relationship between the price paid for the target and the bidder's stock value.

Prior literature extensively analyses the implication of the method of payment on managers' M&A decisions. In particular, Shleifer and Vishny (2003) propose a theoretical model, where managers are completely rational, understand stock market inefficiencies, and take advantage of them, in part

[2] As detailed by Erickson and Wang (1999), the target usually hires an investment bank to evaluate the exchange ratio. The investment bank evaluates a variety of information, including earnings and the content of corporate disclosures, in arriving at their opinion of the fair market (appraised) value of the acquiring firm.

through merger decisions. They suggest that M&As are a form of arbitrage by rational managers operating in inefficient markets. Assuming managers respond rationally to irrational markets, the findings indicate that overvalued firms are more likely to become acquirers, while undervalued firms are more likely to become takeover targets. Consistent with the model by Shleifer and Vishny (2003), Rhodes-Kropf and Viswanathan (2004) empirically demonstrate that overvalued stocks increase the likelihood of mergers occurring, and these mergers are more likely to be financed with stock. In addition, Rhodes-Kropf, Robinson, and Viswanathan (2005), Ang and Cheng (2006) and Dong, Hirshleifer, Richardson, and Teoh (2006) find that merger waves are more likely to occur when market-to-book ratios are high. In particular, Chidambaran, John, Shangguan, and Vasudevan (2010) and Erel, Liao, and Weisbach (2012) evidence that, in order to reduce the target's relative cost, bidders tend to use stock financing when their stock is most overvalued.

This prior evidence shows that managers tend to strategically time the moment to acquire another firm when the stock is most valuable. It also implies that managers are likely to resort to various strategies to induce such misvaluation. Shleifer and Vishny (2003) argue that this phenomenon prevalence is becoming increasingly apparent, as evidenced by the number of studies investigating earnings management around stock-for-stock M&As transactions. For instance, Erickson and Wang (1999) study the earnings of acquiring firms in the quarters before the announcement and agreement dates when the acquisitions are paid with acquirers' own stock. Their results show that acquiring firms inflate earnings in the quarter before the deal announcement. In fact, they find that the unexpected accruals are significantly higher by 2 to 3% before the event. They also find that the degree of earnings management is positively related to the size of the deal as a large deal provide stronger incentives to manipulate earnings. Similar evidence is provided by Louis (2004), Higgins (2013), Gong, Louis, and Sun (2008) for lawsuits in the US and Botsari and Meeks (2008) for the London Stock Exchange. In addition to earnings management, He et al. (2020) show how managers of stock-for-stock M&As manage down analyst earnings forecasts prior to announcement.

## 2.2. Stock-for-stock M&As and the manipulation of corporate disclosures

Instead of investigating the manipulation of quantitative financial information, we examine whether M&A deals lead managers to manipulate the contentof the narrative sections contained in corporate disclosures and test whether stock-for-stock acquirers, manipulate this content, relative to cash and hybrid acquirors. This focus on corporate narratives over numbers is supported by strong evidence that stakeholders use qualitative information from corporate disclosures to understand managers' private insights into the firm's financial performance. For example, Davis et al. (2012) highlight the information value of earnings press releases, documenting a positive correlation between the tone of these releases and future firm profitability, as well as investor reactions at earnings announcements. Henry (2008) also finds that the tone of earnings press releases is positively correlated with short-term returns around the disclosure date, even after accounting for a firm's financial data and earnings surprises. Furthermore, Boudt, Thewissen, and Torsin (2018) show that the tone of earnings press releases holds informational value for stakeholders, with its significance increasing with the level of information asymmetry between firms and investors. Overall, the literature strongly supports the idea that investors respond positively to the tone of earnings press releases around earnings announcements.

This focus on corporate narratives instead of numbers is justified by the fact that there is strong evidence that stakeholders use qualitative information from corporate disclosures to infer the manager's private information about the firm'sfinancial performance. For instance, Henry (2008) shows that the tone of earnings press release is positively correlated with short window contemporaneous returns around the date of thedisclosures, even after controlling for a firm's financial information and earnings surprise. In addition, Davis et al. (2012) examine the information value of earnings press releases and document a positivecorrelation between the tone of earnings press releases and future firm profitability, as well as a positive relationship between tone and investors' reaction at earnings announcement. Boudt, Thewissen, and Torsin (2018) further find that the tone of earnings press releases provides information value to stakeholders and that the informativeness of tone increases with the information asymmetry between firms and investors. Similarly, Fu et al. (2021) investigate how the disclosure tone of earnings conference calls helps predict future stock price crash risk. Their result shows how firms exhibiting more pessimistic tone during the current year-end call experience higher stock price crash risk in the coming year. Overall, the evidence in the literature lends considerable support to the notion that investors respond positively to the tone of earnings press releases around the earnings announcement.

However, unlike quantitative information, the narrative sections of earnings press releases remain unregulated, granting managers considerable latitude in how information is presented to stakeholders. This lack of regulation makes the information in these disclosures harder to legally contest and allows managers the opportunity to subtly shape investors' perceptions of the firm's performance by carefully choosing the language used in earnings press releases. There is growing evidence showing that the tone of earnings press releases is also affected by the presence of manager-specific incentives, such as the CEO's compensation structure (Arslan-Ayaydin et al., 2016), the presence of labor unions (Arslan-Ayaydin et al., 2020), the issue of equity and stock option grants (Huang, Teoh, and Zhang, 2014) and CEO narcissism (Marquez-Illescas, Zebedee, and Zhou, 2019). Other studies also corroborate that managers tend to hype their stock by increasing discretionary and optimistic disclosure with the objective of reducing the cost of capital or influencing short-term prices, especially before new stock issues (Huang et al., 2014). This evidence clearly shows that managers not only manipulate earnings numbers, but also engage in the manipulation of qualitative information due the incentives.

Given the role played by disclosures' tone on the firm's stock price, we therefore hypothesize that stock-for-stock acquirers may exploit the content of corporate disclosures and use more optimistic language in their earnings press releases before the M&A announcement. In doing so, managers will hype their stock before the transaction announcement and reduce the cost of the acquisition. We therefore define to the following hypothesis:

*H1a: Stock-for-stock acquirers inflate the tone of corporate disclosures before merger & acquisition announcements.*

However, it is crucial to recognize that opportunistic departures from standard practices aimed at inflating the bidder stock's price can also come with significant costs. One such cost is the loss of credibility (Hirst, Koonce, and Miller, 1999). For example, Mercer (2004) illustrates how investors place less reliance on corporate disclosures when managers are incentivized to mislead or provide inaccurate information during disclosures. During the negotiation of the exchange ratio of the stock-for-stock merger, investors may therefore grow skeptical of the credibility of financial narratives and are therefore likely to discount the information value of the tone of the qualitative information (Arslan-Ayaydin et al., 2016; El-Haj, Rayson, Walker, Young, and Simaki, 2019). As a result, if voluntary corporate narratives are interpreted as a tool to manage investors' impression, with no real value, it is unlikely that the bidders would alter the wording to manage investors' impression of the stock's value before the M&A.

In addition to a loss in credibility, litigation costs also constitute an important risk for bidders before stock-for-stock M&As. There is strong evidence that managers' use of positive words in corporate disclosures significantly increases litigation risk. For example, Rogers, Van Buskirk, and Zechman (2011) observe a positive relationship between managers' use of optimistic language in corporate disclosures and litigation risk. Their research indicates that plaintiffs often target optimistic statements in lawsuits, and firms facing litigation tend to have a disproportionately high number of positive words in their disclosures. This risk of litigation after M&As, which can be highly publicized, can reduce managers' incentives to change the tone of corporate disclosures before the event. Finally, the target's manager is typically an informed rational agent. As a result, they are likely to be familiar with the impression management methods used by the bidder. One could therefore expect rational target managers to adjust their information set during the negotiation of the exchange ratio or to oppose and reject the transaction.

*H1b: Stock-for-stock acquirers do not inflate the tone of corporate disclosures around merger & acquisition announcements.*

### 2.3. Economic benefits of tone management – Deal value, profitability and control mechanisms

The impact of stock-for-stock M&As on tone is likely to vary cross-sectionally with deal- and firm-specific characteristics. We predict that stock-for-stock acquisitions have the largest impact on tone management when the economic benefits at stake for the acquirer are the highest. In particular, given the cost of tone management, we expect tone management to be an increasing function of the size of the transaction value. If the size of the transaction is relatively small, the economic benefit from increasing the stock price would also be relatively small. On the contrary, if the size of the transaction is relatively large, the economic benefits at stake are of greater magnitude, leading bidders to have more incentives to opportunistically increase the tone of the corporate disclosure. As a result, we expect the acquiring firm's tone management before the transaction to be an increasing function of the deal value.

*H2: The management of tone before stock-for-stock M&As increases with the merger's deal value.*

Similar to deal characteristics, we further expect firm-specific characteristics to affect the relationship between tone management and stock-for-stock acquisitions. We expect profitable firms to be less likely to engage in acquisition transactions as they have a high value from their existing operations (Park, Yoon, and Kim, 2013), and therefore already benefit from a relatively high stock price. On the contrary, unprofitable firms are pro-actively looking for targets and more likely to enter acquisition transactions. This relationship is in line with prospect theory, which argues that managers in less profitable firms are more likely to look for high-risk ventures, whereas highly profitable acquirerstend to be more risk averse and to enter M&A deals of higher quality. As a result, we hypothesize that the incentive to manage the tone of corporate disclosures before M&A announcements is stronger for firms with a low past profitability.

*H3: The management of tone before M&As decreases with the acquirer's past profitability.*

Finally, firm-specific monitoring mechanisms are expected to affect the impact of stock-for-stock decisions on tone management. On the one hand, the presence of control mechanisms, such as financial analysts or leverage, have been shown to reduce managerial opportunistic behavior, such as earnings management by providing additional monitoring. For example, Hong, Huseynov, and Zhang (2014) show that the number of analysts following a firm has important monitoring effects on managerial behavior, which results in lower levels of both accrual- and real-based earnings management. Similarly, other studies finds that there exists a negative relationship between leverage and earnings manipulation

(Rodriguez-Perez and van Hemmen, 2010; Alsharairi, 2012), as managers in leveraged firms may face more control from creditors, limiting the manipulation of earnings. On the other hand, the presence of financial analysts may put more pressure on managers and lead them to engage in greater tone management. For instance, Bratten, Payne, and Thomas (2016) show that managers tend to resort to opportunistic behavior when followed by a larger number of analysts. As such, higher analyst coverage increases the scrutiny of managers and leads to manipulative behavior. Other studies also show that firm leverage can be positively associated with earnings management (DeFond and Jiambalvo, 1994; Iatridis and Kadorinis, 2009). Overall, the expected impact of leverage or analysts' coverage on tone management around M&A transactions is ambiguous and remains to be empirically investigated.

*H4a: Tone manipulation decreases with the number of analysts (leverage) following the firm before the stock-for-stock acquisition.*

*H4b: Tone manipulation increases with the number of analysts (leverage) following the firm before the stock-for-stock acquisition.*

## 3. Sample selection and variable definition

In this section, we provide an overview of the model construction, variable definitions and sample construction.

### *3.1. Collection of earnings press releases*

We use earnings press releases as the disclosure because a significant amount of prior research documents the economic relevance of earnings announcements as a means by which firms communicate with investors (see, e.g., Kothari, 2001) and provides evidence that earnings announcements generally lead to a higher market reaction than other SEC filings (Li and Ramesh, 2009). Furthermore, the type of information disclosed in earnings announcements is not as strictly specified by the SEC relative to more highly regulated documents, such as annual reports. As such, earnings press releases leave managers to freely express linguistic and stylistic content as they see fit, particularly around M&As.

We retrieve quarterly earnings press releases from the EDGAR website. Since 2003, publicly listed firms are required to submit earnings releases to the SEC via a Form 8-K within four days of their announcement. We identify earnings releases by selecting 8-K filings containing Item Code 2.02 and exclude those filed beyond the mandated four trading days.

Earnings press releases are located under the tag "<TYPE> EX 99.1". To ensure our sample exclusively comprise earnings press releases, we review all electronic files larger than 10 kilobytes, filtering out conference call announcements and other non-earnings-related notices. We manually reformat the downloaded HTML files containing earnings press releases into a readable format. While parsing the text, we proceed with the following steps applied in prior research (e.g., Boudt et al., 2018; Davis and Tama-Sweet, 2012; Davis et al., 2012; Henry, 2008; Arslan-Ayaydin et al., 2016; Loughran and McDonald, 2011;):

(1) Delete graphics and other infographics.

(2) Re-encoding characters such as   (blank space) or &AMP (&) back to their original ACSII form.

(3) Delete all text appearing within $< TABLE >$ HTML tags, where more than 15% of the nonblank characters are numbers.

Following Davis et al. (2012), we define a minimum threshold of words in the document. We therefore require a press release to have at least 100 words and remove any observation otherwise.

### *3.2. Research design*

To test Hypothesis 1, we estimate the following cross-sectional regression:

$$Tone_{M\&A} = \alpha + \delta \times StockforStock + \beta \times Controls + \varepsilon, \quad (1)$$

where $\varepsilon$ are Newey West standard errors, corrected for heteroskedasticity and autocorrelation.[3] $Tone_{M\&A}$ is the sum of the bidder tone over four earnings press releases prior to the merger

[3] Our main results continue to hold if we (i) cluster at the firm level, (ii) adopt a two-way clustering (firm and year quarter, and two-digit SIC industry and year quarter) (Gow, Ormazaba, and Taylor, 2010; Petersen, 2009), (iii) use the Huber White standard error without clustering or (iv) cluster at the two-digit SIC industry level.

announcement. $StockforStock$ is a dummy variable equal to one for stock deals, and zero for hybrid and cash deals. $\delta$ is the coefficient of interest, which captures the impact of stock-for-stock deals on the tone of earnings press releases. If firms strategically inflate the tone of earnings press releases before stock-for-stock announcements, we expect $\delta$ to be positive.

We define the tone of earnings press releases (in %) as the relative spread between the number of positive and negative words (see, e.g., Brockman and Cicon, 2013; Arslan-Ayaydin et al., 2016; Marquez-Illescas et al., 2019; Fu et al., 2021):

$$Tone_{M\&A} = 100 \times \sum_{q-1}^{-5} \frac{PositiveWords_{\ q} - NegativeWords_{\ q}}{TotalWords_{\ q}}, \tag{2}$$

where $TotalWords$ is the total number of words in the disclosure from firm $i$ in quarter $q$ of year $t$. $PositiveWords$ and $NegativeWords$ are the numbers of positive and negative words in the earnings press releases, respectively. The main issue with this bag of words approach is that there currently exists no consensus in the literature regarding which word list is more appropriate for the analysis of language in contexts such as financial disclosures. One exception prevails, however. Henry and Leone (2016) present evidence that word-frequency tone measures based on domain-specific wordlists (compared to general wordlists, e.g., DICTION) better predict the market reaction to earnings announcements, have greater statistical power in short-window event studies, and exhibit more economically consistent post-announcement drift. We therefore conduct our main analyses based on the Henry (2008) wordlists.[4]

To analyze the impact of stock-for-stock M&As on the tone of earnings press releases before the M&A announcement, we rely on prior literature (He et al., 2020; Erickson and Wang, 1999) and use of the sum of the tone over the five quarters preceding the quarter of the M&A announcement (e.g. the sum of $Tone_{M\&A}$ from $q-5$ to $q-1$). We then take the percentage of that value. The reasoning for this five-quarter period prior to the M&A announcement is that acquirors' merger discussions with the target typically take place over several quarters prior to the formal merger announcement (see, e.g., He et al.,

[4] The Loughran and McDonald (2011) libraries are also a popular choice in the analysis of corporate disclosures. We therefore also provide robustness tests with those libraries. We show in Appendix A that our results remain qualitatively similar if we use Loughran and McDonald (2011) lists to identify positive and negative words.

2020; Erickson and Wang, 1999; Boone and Mulherin, 2008). In spite of negotiation processes varying widely across deals, bidders usually start reaching out to the target three to five quarters before the M&A announcement (He et al., 2020). They then submit formal takeover proposals several months before the M&A announcement. As a result, bidders' incentives for engaging in tone management is likely to become obvious as early as one year before the announcement, but we expect this effect to become stronger during the last two or three quarters before the announcement, as the bidder and target reach the final terms of their agreement. To capture such effect on tone, we therefore take the sum of the tone over the last four semesters.[5]

$Controls$ represent the set of determinants that have been shown to explain the tone of press releases and are measured one year prior to the M&A announcement. Prior literature shows that tone is also related to a set of economic attributes, such as earnings uncertainty, firm characteristics and the firm's informational environment (see, e.g., Henry, 2008; Huang et al., 2014). To control for firm-specific characteristics, we include firm size (measured as the logarithm of the firm's market capitalization, $Size$), leverage ($Leverage$) as the long-term debt divided by total assets at the end of year $t-2$ and the firm's book-to-market ($BTM$) as the ratio of the market value of common equity atthe end of year $t-1$ to the book value of equity in year $t-1$. In line with Arslan-Ayaydin et al. (2016, 2020), we also include four variables to control for the firm's performance: (i) the return on assets in year $t-1$ ($ROA$), which is defined as the ratio of the income before extraordinary items of firm divided by the firm's total assets at the end of year $t-2$, (ii) a dummy variable $Loss$ that equals one if the firm's $ROA$ in year $t-1$ is negative, and zero otherwise, (iii) analysts' forecast error as the forecast error of the previous year ($FE$), and (iv) the 12-month buy-and-hold monthly returns, ending in year $t-1$ ($Return$). To control for the firm's information environment, we further include the logarithm of the number of financial analysts following the firm in year $t-1$ ($NoA$), the dispersion of analysts' forecasts in year $t-1$ ($\sigma_{FE}$) as defined by Das, Levine, and Sivaramakrishnan (1998) and the standard deviation of the firm's profitability ($\sigma_{ROA}$) over the four quarters in the preceding year $t-1$, as defined by Core and

[5] Taking the last two or three semesters before the M&A announcement does not alter our conclusions. The results are available upon request.

Wayne (2002).

We also include an array of deal-specific variables. We first include the logarithm of the number of bidders ($NbrBidders$) competing for the target. We also control for logarithm of the deal value ($DealValue$), which is the transaction's value reported by SDC in millions of dollars. We further add a dummy variable for tender offers ($Tender$), which is defined as an indicator variable that equals one if the SDC reports that the deal is a tender offer, and equals zero otherwise. Furthermore, we control for diversifying deals ($Diversification$), which is an indicator variable that equals one if the bidder is not from the same two-digit SIC industry as the target firm, zero otherwise. Finally, we control for industry, year and quarter fixed-effects. The industry fixed-effects (based on two-digit SIC codes) and time fixed-effects allow our model to control for the general state of the economy, as well as industry- and time-specific characteristics.

To test Hypotheses 2, 3 and 4, we estimate the following model:

$$Tone_{M\&A} = \alpha + \delta \times StockforStock \times Factor + \beta \cdot Controls + \varepsilon, \quad (3)$$

We test for the significance of the coefficients using Newey-West standard errors, where $Factor$ is the set of firm- and deal-specific factors that influence the effect of stock-for-stock M&As on managers' use of tone inflation. For Hypothesis 2, we use the deal value ($DealValue$) provided by the SDC database and predict $\rho$ to be positive. Hypothesis 3 focuses on the influence of firm past performance on the relationship between stock-for-stock M&As and tone management. To proxy for firm performance, we first use the firm's return on assets in year $t-1$ ($ROA$). We then use the firms' stock performance in the year preceding the M&A announcement ($Return$). We predict $\delta$ to be negative for both performance proxies. Hypothesis 4 investigates the impact of firm monitoring on tone management around M&As. We use analysts' coverage ($NoA$) and the firm's leverage ratio ($Leverage$) as proxies for firm monitoring. If analysts and leverage increase the control mechanisms in the firm, we expect $\delta$ to be negative (Hypothesis 4a). On the contrary, if analysts and leverage put pressure on managers and leads them to increase the tone, we then expect $\delta$ to be positive (Hypothesis 4b).

### 3.3. Sample selection

The sample includes 5,558 M&As by U.S. public bidders from 2004 and 2020, retrieved from the Securities Data Company's (SDC's) Mergers and Acquisitions database. We start with 147,188 U.S. domestic M&As with announcement dates between January $1^{st}$, 2004 and December $31^{st}$, 2020. We then follow Kisgen (2009) and require a deal to have a transaction value above $5 million, which leaves us with 98,000 deals.[6] We further require the bidders to be publicly traded firms, which leads to 75,000 deals. Moreover, we keep the deals for which we have available information about the target's ownership status (e.g., public, private or subsidiary) and deal status (completed, withdrawn). We also request the bidder to acquire at least 50% of the target's stocks. This data selection leaves us with 10,456 deals. We then require the bidder to have stock price information in CRSP, accounting datain COMPUSTAT and analyst forecasts in I/B/E/S.

We then keep observations for which we have earnings press releases for five quarters before the quarter that announces the M&A, the quarter of the M&A announcement, as well as for four earnings press releases after the earnings announcement.[7] This restriction significantly decreases the sample size and gives a final number of 5,558 deals, split between 302 stock deals, 4,088 hybrid deals and 1,168 cash deals. Panel A of Table 1 reports the distribution of the M&A sample. Consistent with market conditions, a large proportion of the deals are taking place in the second period of the sample, between 2011 and 2018.

### 3.4. Summary statistics

Panel B of Table 1 provides the deal and firm characteristics for our sample. To address concerns for outliers, we have winsorized all continuous variables in our analyses at the $1^{st}$ and $99^{th}$ percentile. We find that the average deal has a deal value of $65 million, with the bidder having an average market size

[6] Using other cutoffs for the minimum deal size does not affect our results, which are available upon request.
[7] This represents a total of ten earnings press releases per deal.

($Size$) of \$20.783 billion. The data indicate that the acquiring firms are on average profitable with a positive earnings surprise ($FE$), a return on assets ($ROA$) of 1.1% and a positive stock market (*Return*) of 8.6% over the past year. The average bidder also has an average of 9.592 analysts' following its stock and 24% leverage ratio.

< Insert Table 1 here.>

Panel C of Table 1 reports the firm and deal characteristics for deals using different methods of payment. Compared to cash deals, stock deals tend to be less profitable ($ROA$), smaller ($Size$), less leveraged ($Leverage$) and more growth oriented ($BTM$), which is consistent with prior literature that on average stock bidders have a lower book-to-market ratio (Martin, 1996). In addition, stock bids are less likely to be tenders ($Tender$), in competition with other companies ($NbrBidders$) and tend to occur in the same industry ($Diversification$). Furthermore, the deal size ($DealValue$) for stock bidders (\$1.342 billion) tends to be higher than for cash bidders (\$1.001 billion).

Table 2 reports the Pearson and Spearman correlations of our independent variables and relevant control variables. There is a positive correlation between $StockforStock$ and $Tone_{M\&A}$. This positive correlation is in line with Hypothesis 1a, which argues that the tone of earnings press releases is more optimistic before stock-for-stock M&A announcements. In addition, control variables are significantly correlated with $Tone_{M\&A}$, indicating that multivariate regressions are appropriate.

< Insert Table 2 here.>

## 4. Results

This section first describes the univariate relationship between tone and M&A announcements by highlighting the dynamics of tone around stock-for-stock M&As as well as for cash and hybrid deals. Given the strong correlation between tone and the control variables, we then provide the results of the multivariate regressions and show that our results hold after we correct for endogeneity concerns. Finally, we conduct cross-sectional analyses that test whether this manipulation of tone varies across firm and across deals. In particular, we focus on the (i) deal size, (ii) firm performance and (iii) firm monitoring.

### *4.1. Tone management around M&A announcements*

Table 3 provides the average tone for acquirors' five quarterly earnings announcements from Quarter -5 to -1, where Quarter -1 is the bidder's quarterly earnings announcement before the quarter of the M&A announcement. To provide a complete description of the dynamics of tone around the M&A announcement, we also show the average tone for the four quarters following the M&A announcement quarter. The timeline of the evolution of tone around M&A announcements is plotted in Figure 1.

< Insert Figure 1 here.>

For stock-for-stock transactions, Figure 1 shows that both the mean and median tone substantially increase before the M&A quarter. The median tone reaches a peak of 1.460 in Quarter -2, compared to a tone of 1.266 in Quarter -5. This increase means that the tone of earnings press releases of stock bidders is 15.32% higher two quarters before the M&A announcement, compared to five quarters before the announcement, which is both statistically and economically significant. In contrast, we find that the tone of earnings press releases of cash deals does not significantly increase between Quarter -5 and -1. The median tone of cash deals ranges between a minimum of 1.196 in Quarter -3 and 1.260 in Quarter 0. Moreover, the results of *t*-tests and Wilcoxon tests show that, while the tone for stock deals are insignificantly different from cash deals in Quarter -5, the difference progressively becomes more significant as the M&A announcement approaches and reaches a peak in Quarter -2. The difference in the tone of stock and cash deals then decreases and becomes insignificantly different from cash deals again in Quarter 0. This significant difference between stock and cash deals is consistent with the stronger incentives for tone management by stock-for-stock bidders than cash acquirors. This incentive is also clearly evidenced by the tone dynamics of hybrid M&A deals around the M&A announcement. We find that the tone of hybrid deals tends to be more optimistic before the M&A announcement, but at a more moderate degree compared to stock-for-stock deals. This pattern is clearly evidenced in Figure

1.

Furthermore, the stock bidders become less optimistic after the M&A announcement (Quarter 1 to 4). We find that the differences between the stock and cash bidders become statistically insignificant in the quarter of the M&A announcement, as well as in the subsequent ones. The results highlight that bidders showcase less tone inflation after the M&A announcement, where the incentives of manipulating the content of narratives are significantly smaller than before the announcement. Overall, the evidence of Figure 1 and Table 3 provide initial support to Hypothesis 1a that managers tend to manipulate the content of earnings press releases by inflating the tone of earnings press releases prior to the quarterly earnings announcement before a stock-for-stock merger.

< Insert Table 3 here.>

### *4.2. The multivariate setting of tone management*

We next examine the tone of earnings press releases around M&As using multiple regression analyses that control for firm and deal characteristics. Table 4 presents the results of Equation 1, which tests for the impact of stock-for-stock M&As on the tone of earnings press releases around M&A announcements. The dependent variable ($Tone_{M\&A}$) is the average of bidders' tone for five quarters before the M&A announcement (for Quarter -5 to -1). The independent variable of interest is $StockforStock$, which equals one if the M&A deal is stock-based, zero otherwise.[8] In Model (1), the parsimonious specification only includes year and industry fixed effects, but no other control variables. The results show that the coefficient $StockforStock$ is significantly positive, which is consistent with the results in Table 3. Model (2) further includes the control variables. We find that the results remain statistically significant at a 99% confidence interval. In fact, we observe that the magnitude of the coefficient is even larger. Regarding the control variables, a large number of firm- and deal-specific characteristics are correlated with $Tone_{M\&A}$. We find that $Tone_{M\&A}$ is significantly and positively associated with the earnings

[8] $StockforStock$ is defined as a dummy variable that equals one for stock-for-stock deals and zero for hybrid and cash deals. However, excluding hybrid deals and exclusively focusing on cash and stock deals provide qualitatively similar results. Appendix B provides the result description if we exclude hybrid deals, and only select cash and stock deals.

surprise ($FE$), while it is negatively associated with analysts' coverage ($NoA$), the loss dummy ($Loss$) and the standard deviation in past performance ($\sigma_{ROA}$). We also find that the number of competing bidders ($NbrBidders$) significantly increases the tone around the M&A announcement. Finally, if the deal leads to a diversification in the firm's operations ($Diversification$), the tone of earnings press releases is less optimistic.

< Insert Table 4 here.>

To investigate the trends in tone management approaching the M&A announcement, we show in Model (3) to (7) the regressions of quarterly tone for each of the Quarters -5 to -1, respectively. As expected and consistent with our univariate results, we find that the tone is significantly increasing from Quarter -5 to Quarter -2. The coefficient for $StockforStock$ in Model (3) starts at 0.099, which is significant at a 95% confidence level. The coefficient for the same variable increases to 0.128 for Quarter -4, 0.155 for Quarter -3 and 0.176 for Quarter -2. It then decreases to 0.138 for Quarter -1. All are significant at a 99% confidence interval. This evidence highlights that stock bidders' tone in earnings press releases are more optimistic when approaching the public announcement of the M&A. Because Erickson and Wang (1999) find that stock bidders tend to manipulate their earnings numbers before the M&A announcement, it is possible that the use of tone inflation stems from this upward management of earnings, instead of the approaching M&A announcement. To correct for this effect, we control for earnings management by introducing the absolute value of discretionary accruals as a proxy for earnings management ($EarningsManagement$) (see, e.g., Prior, Surroca, and Tribó, 2008). We show in Appendix C that the relationship between $StockforStock$ and tone inflation ($Tone_{M\&A}$) remains highly significant after controlling for earnings management.

While Table 4 shows a significant increase in the tone just before the announcement of stock-for-stock M&As relative to cash deals, it remains unclear whether bidders become more positive or less negative, or both in their communications with stakeholders. We therefore distinguish between positive and negative tone as dependent variables in Equation 1. Panel A and B of Table 5 report the results for the positive and negative tone, respectively. We find that $StockforStock$ is significantly positive with the number of positive words, but is insignificantly associated with negative tone before the M&A announcement. The results suggest that the announcement of stock-for-stock M&As significantly increases the use of positive words in earnings press releases, rather than decreases the use of negative words in the document.

< Insert Table 5 here.>

### *4.3. Endogeneity concerns*

Our results so far provide strong evidence of a manipulation of the content of earnings press releases in the quarters leading to the M&A announcement of stock-for-stock deals. This result highlights the incentive for managers to engage in the inflation of tone to boost the stock price and reduce the target acquisition cost before the M&A announcement. However, obtaining causal evidence is challenging as the decision to use stocks to acquire targets may be based on a variety of attributes that are not readily quantifiable, but that might be correlated with the tone of earnings press releases. For instance, hubris behavior or managerial overconfidence in the firm's future performance may explain the inflation in tone (Moeller, Schlingemann, and Stulz, 2004; Malmendier and Tate, 2008). Given that such variables are not accurately quantifiable, our current economic interpretation may either be a spurious statistical artefact (that is, our significance is a false positive), or driven by an omitted variable bias. Several methodological approaches exist to account for this endogeneity. In line with prior studies (see, e.g., Thewissen, Thewissen, Torsin, and Arslan-Ayaydin, 2022; Colombo and Grilli, 2010), we employ a large array of endogeneity tests to confirm the causality of our results: (i) the restricted control function (rCF), (ii) pseudo-event study, (iii) instrumental variable, (iv) propensity- score matching and (v) entropy balancing.

#### *4.3.1. Propensity-score matching*

One could argue that firms conducting stock-for-stock deals tend to be more profitable, larger and less leveraged (e.g. more cash). If that is the case then it may be that the more optimistic tone is explained by firm-specific characteristics, such as performance or size. As a result, it may not be the method of payment that is leading firms to be more optimistic, but rather that they are more profitable or less leveraged. In addition, our original OLS model assumes a linear relationship between the controls and the dependent variable. If these associations are not linear then our model's specification is mis-specified.

To circumvent such issues, we use the propensity score matching (PSM) method to construct a new sample. Contrary to the OLS regression, which assumes a linear relationship between the dependent variable and the covariates, the PSM does not pre-suppose any functional association. In fact, it is an empirical method adjusted to assess a causal relationship. Many studies in prior literature use this approach to address concerns that the treated sample is not similar to the control sample (Thewissen et al., 2022). As such, this practice helps mitigate the issue that omitted firm-specific characteristics may simultaneously explain both the choice of the method of payment and tone management.

We first divide our sample into two subsamples based on the method of payment. We consider the subsample of firms that use stocks as a method of payment as our treatment group. Our control group contains deals that either use cash or both stock and cash as method of payment. We then run a logit regression on all the firm-specific variables of Equation 4 to estimate the probability that a firm would choose stocks to conduct an M&A deal:

$$\begin{aligned} StockforStock = \; & \alpha + \beta_1 \times ROA + \beta_2 \times BTM + \beta_3 \times Return + \beta_4 \times \log Size \\ & + \beta_5 \times \sigma_{Ret} + \beta_6 \times \sigma_{ROA} + \beta_7 \times Leverage + \beta_8 \cdot\times Loss + \beta_9 \times FE + \beta_{10} \times \sigma_{FE} \\ & + \beta_{11} \times \log NoA + \text{Year and industry fixed effects} + \varepsilon. \end{aligned} \tag{4}$$

We then match each stock-for-stock deal with the closest propensity score to the cash or hybrid deals based on the neighbor matching method. This means that for each observation from the treatment group, we get one observation from the control sample. As a result, our final sample for the PSM second-stage regression consists of 604 observations (302 stock deals and 302 cash or hybrid deals). Panel A of Table 6 depicts differences in mean scores for all covariates between the stock-for-stock firms and the propensity matched control group. Overall, while we observe a significant difference across all variables before the matching procedure, we observe no statistically significant differences in the means between the two groups after we match each deal. Overall, the results in Table 6 suggest that the propensity score matched control group mirrors the stock-for-stock deals reasonably well, given the covariates.

< Insert Table 6 here.>

Panel B reports the differences in the mean and median tone between Quarter -5 and 0. Similar to our initial results, we find that the tone of stock-for-stock deals significantly increases as the date of the M&A announcement approaches. The difference between the mean tone for the treatment and control groups becomes significantly different from Quarter -4 to -1, while the difference in the median tone is significant between Quarter -3 and -2. This statistical difference is confirmed in Model (2) of Panel C. The coefficient *StockforStock* is positive and statistically significant at a 95% confidence interval.

### *4.3.2. Entropy balancing*

In spite of the PSM approach being commonly used in prior literature (Shipman, Swanquist, and Whited, 2017), more recent literature discusses that PSM is subject to several caveats. First, the PSM often relies on random matches and fails to meet the covariate balance condition (King and Nielsen, 2016). In addition, the nearest neighbor method ignores a subset of control observations, which can cause some loss of information or lack of power (Hainmueller, 2012). Contrarily to the PSM, entropy balancing almost always reaches a high covariate balance. It reweights each covariate observation through an iterative process until the first, second and even higher moments of the control group equal those of the treated group.

To conduct the entropy balancing, we use the same matching variables used in the PSM. We then match the mean, variance and skewness of the deals. After multiple iterations, each control observation is assigned a weight and we use these weights to estimate the regressions. As shown in Panel A of Table 7, there is convergence for the mean, variance and skewness of the control variables. Panel B shows that the tone difference between the treatment and control groups become significantly larger as the date approaches the M&A announcement. This trend is verified in Model 1 of Panel C, with a positive and significant coefficient of 0.145 for the variable $StockforStock$.

< Insert Table 7 here.>

### *4.3.3. Instrumental variable approach*

The instrumental variable approach ensures that unobserved features that may simultaneously explain the method of payment and the tone of earnings press releases are not driving our results. A proper instrument exclusively affects the outcome variable ($Tone_{M\&A}$) through its impact on the independent variable ($StockforStock$). As such, it needs to fit the relevance (e.g. the instrument should be correlated with $StockforStock$) and exclusion criteria (no direct correlation between the instrument and $Tone_{M\&A}$).

We define our instrument as the percentage ownership of institutional shareholders as of the month prior to the acquisition announcement date (*Inst*). Institutional investors are known to actively conduct costly monitoring and to take action that more properly align managers' interests with those of shareholders (Black, 1992). For instance, institutional shareholders are in a position to promote an appropriate management compensation structure, strengthen the institutional voice on the board of directors and serve on the board itself (Martin, 1996). In addition, some institutional investors directly contact senior management and thus may influence the conditions of the acquisition bid. As empirical evidence shows that stock-based deals typically reduce the wealth of the acquiring firm, the likelihood of acquisitions being financed with stock should be lower when institutional shareholders are higher. In line with this reasoning, Martin (1996) provide strong evidence of a negative relationship between the proportion of institutional ownership and stocks as the method of payment.

Our correlation analysis confirms Martin's results. The correlation between our instrument, *Inst*, and the independent variable, *StockforStock*, is highly significant and negative (r(5740) = –8%, $p<0.001$). On the other hand, we find no significant correlation between *Inst* and *Tone* (r(5740) = 1%; p = 0.516). Given that our instrument fits the restriction and exclusion criteria, we conduct the following regression:

$$StockforStock = \alpha + \zeta \cdot \text{Inst} + \beta_1 \times ROA + \beta_2 \times BTM + \beta_3 \times Return + \beta_4 \times \log Size \quad (5)$$

$$+ \beta_5 \times \sigma_{Ret} + \beta_6 \times \sigma_{ROA} + \beta_7 \times Leverage + \beta_8 \cdot\times Loss + \beta_9 \times FE + \beta_{10} \times \sigma_{FE}$$

$$+ \beta_{11} \times \log NoA + \text{Year and industry fixed effects} + \varepsilon.$$

where $\varepsilon$ are Newey West standard errors, corrected for heteroskedasticity and autocorrelation. We report this result in Model (1) of Table 8. As expected, the coefficient for the variable $Inst$ is negative (-0.637) and statistically significant at a 95% confidence level. In addition, the IV diagnostic tests confirm the choice of $Inst$ as instrumental variable. The F-statistic testing whether instruments are unrelated to the endogeneous regressor indicates that our instrumental variable is sufficiently strong.

< Insert Table 8 here.>

We next obtain the fitted values from Equation 5 and repeat our main analysis using the fitted observations. The result is provided in Model (2) of Table 8. We find a statistically significant and positive coefficient between $Tone_{M\&A}$ and $StockforStock(fitted)$. In addition, the Wu-Hausman statistic shows that our OLS estimation is an equally consistent method to the IV analysis, which increases our confidence in the outcome of our initial analyses.

*4.3.4. Restricted control function*

As a fourth method to correct for endogeneity concerns, we rely on seminal work by Heckman (1978) and reconduct our analyses using a two-step restricted control function (rCF) regression. The rCF approach reduces the possibility of endogeneity generated by spurious correlations or reverse causality, leading to the "experimental average treatment effect" (Heckman, 1978; Colombo and Grilli, 2010; Thewissen, Shrestha, Torsin, and Pastwa, 2022) of the method of payment ($StockforStock$) on the tone of earnings press releases ($Tone_{M\&A}$). The benefit of the rCF approach over matching method is that, rather than assuming that the conditioning set of relevant control variables is sufficiently complete, the rCF method models omitted variables (Heckman and Navarro-Lozano, 2004). Another feature is that the generalized residual obtain from the first stage can be regarded as an explicit test for endogeneity (Colombo and Grilli, 2010).

In the first step of the restricted control function analysis, we estimate the probability that the deal will use stocks as a method of payment, based on the same control variables as those employed in Equation 1. Using this model, we calculate the generalized residual ($GENRES$) and insert this residual as an additional control variable in a second step regression, where we estimate the impact of the method of payment on the tone. We report the results of the first-step regression in Model (3) and the results of the second step in Model (4) of Table 8 (Panel B). We find that the relationship between $Tone_{M\&A}$ and $StockforStock$ remains significantly positive.

*4.3.5. Pseudo-event study*

We then conduct pseudo-event analysis by shifting the merger announcement dates forward and backwards by one year (e.g. four quarters). Table 9 provides the results of Equation 1 using the two pseudo- events. Model (1) and (2) present the results when we shift the M&A announcement four quarters backward and forward, respectively. We find that in both situations, the estimate forthe $StockforStock$ variable is statistically insignificant, which strongly departs from the significantly positive coefficients obtained in Table 4.

< Insert Table 9 here.>

### 4.4. Factors moderating and reinforcing the effect of stock-for-stock M&A announcements

In the previous section, we establish a clear causal relationship between stock-for-stock deals and managers' manipulation of the tone of earnings press releases. However, the impact of such transactions is likely to vary cross-sectionally with deal- and firm-specific characteristics. In this section,we conduct several cross-sectional analyses that test whether this behavior changes across deals and firms. In particular, we focus on (i) the deal size, (ii) the firm's past performance and (iii) monitoring. The results of Equation 3 are reported in Table 10.

< Insert Table 10 here.>

### 4.4.1. Deal size

Hypothesis 2 examines the impact of the deal's size on the propensity to manage the content of corporate disclosures. Since tone management is not costless, when the economic benefits are small, there is a lower incentive to manipulate the tone of corporate disclosures. On the contrary, if the economic benefits at stake are large, the incentive to inflate the tone of earnings press releases increases before the M&A announcement. In line with this reasoning, we run Equation 3 and report the results in Model (1) of Panel A in Table 10. Overall, our results are consistent with this hypothesis. We find that the interaction coefficient $DealValue \times StockforStock$ is positive and significant at a 99% confidence level. This positive coefficient indicates that greater the deal's value, the larger is the manager's incentiveto inflate the tone of earnings press releases before stock-for-stock acquisitions. With a magnitude of 0.218, the coefficient for the $StockforStock$ variable remains significantly positive and even increases in magnitude compared to the coefficient of 0.152 in Table 4. Overall, our results provide support to the hypothesis that tone inflation is positively related to economic benefits at stake in the merger.

### 4.4.2. Firm performance

Hypothesis 3 predicts that tone management before stock-for-stock M&As decreases with the acquirer's profitability. In Panel B of Table 10, Model (2) first tests the moderating impact of the acquirer's return on assets ($ROA$). We find a negative and significant effect for the interaction variable $ROA \times StockforStock$. In Model (3), we observe a similar relationship if we consider $Return$ as a proxy for the acquirer's past stock performance. However, while the main effect of $StockforStock$ remains positive and highly significant for Model (3), the coefficient for $StockforStock$ is insignificant for Model (2), which suggests that the tone inflation before stock-for-stock M&As tends to be concentrated among firms that suffer from a low $ROA$. Taken together, these results are in line with Hypothesis 3 and indicate that firm past profitability tends to significantly decrease managers' use of tone inflation before stock-for-stock M&A transactions.

#### 4.4.3. Firm monitoring

Hypothesis 4 further tests the impact of monitoring mechanisms on the management's likelihood in manipulating the tone of earnings press releases before stock-for-stock M&A transactions. While Hypothesis 4a predicts that analysts and leverage act as monitoring mechanisms that limit managerial opportunistic behavior, Hypothesis 4b argues that analysts' coverage and leverage tend to increase the pressure on managers to meet targets, which leads them to resort to opportunistic behavior. In Panel C of Table 10, we first test the impact of analysts' forecasts ($NoA$). In Model (4), we find a negative and significant effect for the interaction variable $StockforStock \times NoA$. In line with this, we also find in Model (5) a significantly negative interaction effect between $StockforStock$ and $Leverage$. Taken together, these results provide support to Hypothesis 4a and indicate that analysts and leverage act as monitoring mechanisms that tend to limit managerial opportunistic behavior before stock-for-stock M&A transactions.

## 5. Additional analyses – Stock market reaction and the target's behavior

Based on the presence of tone inflation before M&A announcements, it is central to better understand whether and how the market responds to such strategic manipulation of the qualitative content of corporate disclosures. In this section, we are interested in quantifying the dollar impact of tone inflation in boosting the stock price, as well as in studying the relationship between tone inflation and the bidder's post-merger stock return in the long run. We further analyze the target's behavior around the M&A announcement.

### *5.1. Does tone management profit bidders?*

#### 5.1.1. The effect of tone management on the bidder's stock price before the M&A announcement

To assess the effect of the acquirer's tone management on its stock price, we conduct regressions of its cumulative abnormal returns around pre-merger earnings announcements using the sample of the acquirer's five quarterly earnings announcements before the merger:

$$CAR[-1;\,+1] \;=\; \alpha + \beta_1 \times AbTone_{M\&A} + \beta_2 \times ExpectedTone_{M\&A} \qquad (6)$$
$$+\beta_3 \times Controls + \varepsilon,$$

where the dependent variable is the cumulative abnormal return ($CAR[-1;\,+1]$) of the acquirer from one day before the earnings announcement of quarter *t* until one day after the earnings announcement of quarter *t*. The main independent variables include the abnormal tone ($AbTone_{M\&A}$) and the *expected* tone ($ExpectedTone_{M\&A}$). The expected tone ($ExpectedTone_{M\&A}$) yields the normal tone, i.e. the tone that is expected when the M&A announcement has no influence on the tone. The expected tone is estimated using the following equation:

$$Tone_{M\&A} = \alpha + \beta \times Controls + \varepsilon. \qquad (7)$$

Following Arslan-Ayaydin et al. (2016), our measure of abnormal tone $AbTone_{M\&A}$ is the difference between the tone of earnings press releases ($Tone_{M\&A}$) and the expected tone ($ExpectedTone_{M\&A}$). By construction, the abnormal level of tone ($AbTone_{M\&A}$) is the residual component ($\varepsilon$) of Equation 7 and designed to be uncorrelated to the firm's fundamentals, information asymmetry or environment. We control for a number of firm characteristics, such as firm performance ($ROA$), stock return standard deviation ($\sigma_{Ret}$), loss dummy ($Loss$), book-to-market ratio ($BTM$), analysts' forecast error ($FE$), the dispersion in analysts' forecasts ($\sigma_{FE}$) and the number of financial analysts ($NoA$). We also control for industry, year and quarter fixed effects. The coefficient $\beta_1$ in Equation (6) measures the impact of a bidder's tone inflation on its stock return, while $\beta_2$ represents the influence of the acquirer's expected tone on its stock return. A significant positive $\beta_1$ indicates a net benefit from tone inflation after controlling for the expected (true) tone in the earnings press release.

Model (1) of Table 11 presents the regression of $CAR[-1;\,+1]$ on $AbTone_{M\&A}$ and $ExpectedTone_{M\&A}$ with the control variables. The coefficient estimate of $\beta_1$ is 0.003 and significant at a 1% confidence level, which indicates that a bidder's pre-merger stock return is increasing with tone inflation. On the contrary, the $ExpectedTone_{M\&A}$ coefficient is insignificantly different from zero, which suggests that the normal tone of earnings press releases has little impact on investors' reaction at earnings announcement. Overall, this result is consistent with Arslan-Ayaydin et al. (2016) and shows how bidders can benefit from inflating the tone of earnings press releases, and thereby increase the stock price.

< Insert Table 11 here.>

We further estimate the dollar savings of acquisition costs due to tone management for an average stock bidder. In Model (1), $\beta_1$ is 0.003, which suggests that the net benefit of tone management in terms of an increase in stock returns around the quarterly earnings announcement is 0.3%. We therefore take all acquirer-quarters for stock bidders in our sample and calculate the dollar saving of acquisition costs for each bidder-quarter as $DealValue \times AbTone_{M\&A} \times$ 0.3%. To estimate the dollar savings for an average stock bidder in our sample, we average the dollar savings across acquirers, quarters and then multiply this average by five as our analysis covers a bidder's five quarterly earnings announcements before the M&A announcement. This approach estimates that the dollar saving of acquisition cost through tone management is around $21,777,798 for an average stock bidder in our sample, which is economically significant. We also conduct a robustness check by using a three day window around the earnings announcement date (CAR[-3;+3]) and find similar results, as shown in Model (2) of Table 11. Overall, the results of Table 11 show that tone manipulation is likely to bring considerable benefits to acquirers before the M&A announcement.

#### *5.1.2. The impact of tone management on the bidder's long-term return after the M&A announcement*

If tone management leads to a temporary increase in the acquirer's stock price, we would expect such mispricing to be subsequently corrected in the long run. For instance, Sloan (1996) finds a significant inverse relation between accruals and long-term returns. Similarly, Arslan-Ayaydin et al. (2016) and

Huang et al. (2014) show how abnormal tone in earnings press releases lead to a temporary increase in stock price, which is subsequently corrected by the market in the long term. They interpret their result as evidence that investors can see through the tone inflation in corporate disclosures and therefore subsequently discount the information signal in the abnormal tone in the presence of managerial opportunistic motives. To examine this possible reversal of stock prices, we estimate regressions of bidders' short term cumulative abnormal returns around the M&A announcement ($CAR[-3; +3]$), as well as the long term cumulative abnormal returns over the 60, 180, 360 trading-day windows after the M&A announcement.

Table 12 regresses the abnormal returns around M&A announcements on its premerger abnormal tone ($AbTone_{M\&A}$) and a dummy variable for stock payment, as well as their interaction term.We also control for various firm- and deal-specific characteristics. Model (1) presents the regression results for the cumulative abnormal returns over the trading window [-3;+3] around the earnings announcement date. Our results show that the coefficient estimate of the interaction variable ($Stockforstock \cdot AbTone_{M\&A}$) is significantly positive, which indicates that pre-merger tone management is associated with a positive stock price reaction around the M&A announcement date.

< Insert Table 12 here.>

However, our results show a clear reversal after the M&A announcement. Model (2), (3) and (4) repeat the regression analyses using [1;+60], [+1;+180] and [+1;+360] trading windows, respectively. We find that the magnitude of the coefficient for the interaction variable $Stockforstock \cdot AbTone_{M\&A}$ progressively decreases with the trading window. While the interaction coefficient remains positive but statistically insignificant for Model (2), it becomes statistically negative for Model (3) at a 90% confidence level for Model (3) and 95% confidence level for Model (3). This negative coefficient shows that premerger tone management is associated with statistically lower bidder returns in the 60-,180- and 360-day period after the M&A announcement. Overall, this evidence shows that, although tone inflation before the M&A announcement temporarily boosts stock prices prior to the deal announcement, investors can progressively see through this tone inflation, which leads this effect to be progressively reversed after the M&A is completed.

### *5.2. The tone management by the target*

The target is likely to be aware of the bidder's incentive to manipulate the tone upwards before the M&A announcement in an attempt to increase its stock price. We therefore analyze the manipulation of tone of target firms around the M&A announcement. We re-estimate Equation 1 using our sample of target firms, based on the same event date and tone definitions as defined above. Our results in Table 13 show that the coefficient for the variable $Stockforstock$ is insignificantly different from zero at traditional confidence levels. Overall, this result tends to indicate that the target does not engage in tone manipulation before M&A announcements, which may be explained by the fact that the target ignores the potential purchase until the bidder initiates the negotiations. Given that the M&A is typically announced and agreed in the same quarter, the target has limited time to manipulate its earnings before the merger. However, note that defining a sample of firms based on the target significantly decreases the sample size to 37 observations, which may limit the inferences we can draw from our analysis on the target firms.

< Insert Table 13 here.>

### *5.3. The structure of earnings press releases before mergers and acquisitions*

The present paper provides ample evidence on the manipulation of earnings press release tone preceding M&A announcements. However, corporate disclosures are structured in a specific way that reflects managers' organization of interconnected ideas and helps users interpret and comprehend their message (Boudt and Thewissen, 2019; Kintsch and Yarbrough, 1982). As such, it is important to recognize that managers not only present individual pieces of information, but also construct a disclosure narrative that conveys their interpretation of current performance and its implications for future firm performance. We therefore examine whether the structure of the narrative also gets strategically manipulated prior to stock-for-stock M&A announcements.

Boudt and Thewissen (2019) and Allee et al. (2015) demonstrate that this structured format of corporate disclosures holds valuable information for investors and significantly influences stock prices. Boudt and Thewissen (2019) show that corporate disclosures exhibit an intra-textual dispersion of sentiment, strategically positioning positive and negative words within the disclosure as a subtle form of impression management. They further find that managers tend to structure corporate disclosures in a way that fosters a more positive perception of the underlying message. Specifically, they uncover a "smile" pattern in the frequency of positive words, and a "half-smile" pattern in the distribution of negative words, with negative words predominantly appearing at the beginning of the letter. Allee et al. (2015) further find that an evenly distributed tone throughout the narrative signifies positive news, while an uneven distribution isolates the news to specific negative performance components. Their findings suggest that the market responds more favorably (adversely) when positive (negative) tone is dispersed. Based on this evidence, we argue that, in addition to tone inflation, bidders also strive to optimize their stock price by manipulating the structure of the narrative content in corporate disclosures. We anticipate that managers increase (decrease) the dispersion of positive (negative) tone in earnings press releases before M&A announcements to enhance investors' reactions and optimize their stock price.

Following Allee et al. (2015) and Boudt and Thewissen (2019), we use the average reduced frequency ($ARF_{M\&A}$) to measure tone dispersion. The disclosure is divided into equally sized sections based on the number of tone words from the Henry (2008) libraries of positive and negative words. The number of sections containing at least one tone word is then counted and divided by the total number of sections, yielding $ARF_{M\&A}$. We follow the same procedure to measure the ARF of the positive (negative) tone, but only use the list of positive (negative) words from the Henry (2008) library. Equation (1) is then rerun, replacing the dependent variable with the disclosure's ARF:

$$ARF_{M\&A} = \alpha + \delta \times StockforStock \times Factor + \beta \cdot Controls + \varepsilon. \quad (8)$$

If bidders attempt to increase the dispersion of the net ($ARF_{M\&A}$) or positive tone ($ARF_{PosToneM\&A}$) before the M&A announcement, a positive $\delta$ coefficient is expected. On the contrary, if bidders attempt to concentrate bad news and, thereby, decrease the dispersion of negative tone ($ARF_{NegToneM\&A}$), a negative δ is expected.

The findings in Column (1) of Table 14 indicate a statistically significant positive association between the $StockforStock$ variable's $\delta$ coefficient and the $ARF_{M\&A}$ measure of tone dispersion, at a 90% confidence level. Further analysis reveals that this increase in tone dispersion around the M&A announcement primarily stems from a larger dispersion in positive tone. Columns (2) and (3) of Table 14 divide the $Tone_{M\&A}$ measure into the Henry (2008) libraries of positive and negative words. While M&A announcements appear to have no impact on the dispersion of negative tone ($ARF_{NegToneM\&A}$) across the earnings press release (Column (3)) with a coefficient of 0.001 for the $StockforStock$ variable, managers tend to increase the dispersion of positive tone ($ARF_{PosToneM\&A}$) before the M&A announcement. Column (2) shows a coefficient of 0.003, which is significant at 90% confidence interval. Overall, this additional analysis aligns with the notion that managers strategically distribute positive news throughout the earnings press release to elicit a more positive response from investors. It suggests that stock-for-stock bidders not only inflate the tone of earnings press releases, but also slightly adjust the structure of corporate disclosures. However, it is important to note that the low significance of this additional analysis implies that the impact of M&A announcements on the dispersion of positive tone is marginal compared to the influence of stock-for-stock M&As on tone and constitute only a second-order effect on the stock price.

## 6. Conclusion

This paper contributes to prior literature by investigating whether stock-for-stock M&As induce managers to strategically manipulate the content of corporate disclosures by inflating the tone of earnings press releases to positively influence the stock price, and therefore reduce the target's acquisition cost. Based on a sample of 5,558 M&A deals between 2004 and 2020, we find evidence that stock bidders are more likely than cash or hybrid bidders to engage in tone management and that the tone of earnings press releases significantly increases by 15% in the year preceding the M&A announcement. This result cannot be explained by differences in information asymmetry, earnings management, industry or time fixed effects and has been tested for endogeneity biases, which strengthens the causal relationship between stock M&As and tone inflation. We provide further insights on the underlying mechanisms by revealing more pronounced effects of stock-for-stock deals on tone inflation for (i) larger M&A deals, (ii) firms with lower past performance and (iii) firms with less control mechanisms. We further find that increasing the tone of earnings press releases is associated with an increase in stock prices in the pre-merger period. In fact, this tone inflation leads to savings in acquisition costs by $21.777 million, on average. However, in the long-term, we document that stock bidders suffer from lower stock returns, which is in line with prior evidence that firms that manipulate earnings to beat expectations are still able to benefit from higher stock returns, but also larger long-term losses. Overall, our paper therefore raises essential ethical concerns with respect to the quality of the narratives contained in corporate disclosures and calls for market agents to consider the impact of the method of payment in M&A deals to enhance their capacity to interpret and process the disclosed information in earnings announcements.

## 7. Appendix

**Appendix A: Robustness Tests - Multivariate Analysis of Bidders' Tone around M&A Announcement Quarters using the Loughran and McDonald (2011) libraries**

| Dependent Variable | Sum of Quarterly $Tone^{LM}_{M\&A}$ | | Bidder's quarterly tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | -0.0004*** | 0.003 | 0.002 | 0.002 | 0.002 | 0.003 | 0.004 |
| | (0.0001) | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) |
| *StockforStock* | 0.043*** | 0.032** | 0.021 | 0.025** | 0.029*** | 0.052*** | 0.020 |
| | (0.011) | (0.009) | (0.011) | (0.005) | (0.004) | (0.005) | (0.012) |
| *ROA* | | -0.001 | -0.001 | -0.001 | -0.004 | 0.002 | -0.001 |
| | | (0.003) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| *Loss* | | -0.002*** | -0.001*** | -0.002*** | -0.002*** | -0.002*** | -0.002*** |
| | | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) |
| *Return* | | -0.004 | -0.003 | -0.001 | -0.002 | 0.001 | 0.0004 |
| | | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| *FE* | | 0.0003** | 0.002 | 0.003** | 0.003* | 0.002 | 0.003** |
| | | (0.0001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| *BTM* | | -0.001*** | -0.001*** | -0.001*** | -0.001*** | -0.001*** | -0.001*** |
| | | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| $Size^{+}$ | | -0.005*** | -0.004*** | -0.005*** | -0.005*** | -0.001*** | -0.004*** |
| | | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| *Leverage* | | -0.001*** | -0.001*** | -0.001*** | -0.001*** | -0.001** | -0.001*** |
| | | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| $\sigma_{Ret}$ | | -0.019*** | -0.020*** | -0.011** | -0.023*** | -0.022*** | -0.021*** |
| | | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| $\sigma_{ROA}$ | | 0.0003 | -0.0001 | -0.0001 | 0.001 | 0.001*** | 0.001*** |
| | | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| $NoA^{+}$ | | -0.003 | -0.001 | -0.001 | -0.004 | 0.001 | -0.001 |
| | | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| $\sigma_{Analyst}$ | | 0.005 | -0.002 | 0.004 | 0.001 | 0.001 | 0.001 |
| | | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) ) |
| $NbrBidders^{+}$ | | 0.004 | 0.001 | 0.002 | 0.001 | -0.002 | -0.001 |
| | | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) |
| *Diversification* | | 0.004** | 0.003* | 0.004** | 0.003** | 0.004** | 0.003* |
| | | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| *Tender* | | 0.004 | 0.005 | 0.001 | 0.001 | 0.001 | 0.0003 |
| | | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| $DealValue^{+}$ | | -0.004 | 0.001 | -0.002 | -0.001 | -0.005 | -0.001 |
| | | (0.004) | (0.005) | (0.005) | (0.005) | (0.005) | (0.004) |
| Year fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.001 | 0.076 | 0.055 | 0.063 | 0.062 | 0.065 | 0.063 |
| Adjusted $R^2$ | -0.001 | 0.069 | 0.048 | 0.055 | 0.055 | 0.058 | 0.055 |

This table provides evidence on Hypothesis 1. It reports the regressions of bidders' tone on the method of payment and other control variables. Instead of measuring tone based on the Henry (2008) libraries, we measure tone based on the Loughran and McDonald (2011) libraries. Stock deals use stocks as the only method of payments, while the other deals either use only cash or both cash and stock as the method of payment. The dependent variable for Model (1) and (2) is the sum of the tone over the five semesters before the semester of the M&A announcement. $StockforStock$ is a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Appendix B: The impact of M&A announcement of stock and cash deals on the tone of earnings press releases, excluding hybrid deals**

| Dependent Variable | Sum of Quarterly $Tone_{M\&A}$ | | Bidder's quarterly tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | 1.287*** | 0.813** | 0.500 | 0.322 | 0.892* | 0.849* | 1.191*** |
| | (0.017) | (0.380) | (0.488) | (0.477) | (0.471) | (0.448) | (0.440) |
| $StockforStock_{NoHybrid}$ | 0.136*** | 0.140*** | 0.132** | 0.145*** | 0.151*** | 0.130** | 0.136*** |
| | (0.037) | (0.043) | (0.056) | (0.054) | (0.054) | (0.051) | (0.050) |
| ROA | | 0.185 | 0.517 | 0.565 | -0.478 | 0.380 | 0.271 |
| | | (0.689) | (0.885) | (0.865) | (0.854) | (0.812) | (0.797) |
| Loss | | -0.030 | 0.005 | 0.034 | -0.081 | -0.020 | -0.054 |
| | | (0.051) | (0.066) | (0.064) | (0.064) | (0.060) | (0.059) |
| Return | | -0.592 | -0.715 | -0.948* | -0.386 | -0.124 | -0.908* |
| | | (0.422) | (0.542) | (0.530) | (0.523) | (0.497) | (0.488) |
| FE | | -0.027 | -0.059 | -0.090 | -0.019 | 0.037 | -0.037 |
| | | (0.094) | (0.121) | (0.118) | (0.117) | (0.111) | (0.109) |
| BTM | | -0.236*** | -0.198** | -0.231*** | -0.252*** | -0.212*** | -0.251*** |
| | | (0.063) | (0.080) | (0.079) | (0.077) | (0.074) | (0.072) |
| Size (log) | | -0.004 | -0.005 | -0.014 | 0.007 | -0.004 | -0.002 |
| | | (0.012) | (0.015) | (0.015) | (0.015) | (0.014) | (0.014) |
| Leverage | | -0.422*** | -0.490*** | -0.484*** | -0.390*** | -0.414*** | -0.401*** |
| | | (0.080) | (0.103) | (0.101) | (0.100) | (0.095) | (0.093) |
| $\sigma_{Ret}$ | | 0.033 | -0.088 | -0.023 | 0.006 | 0.044 | 0.107** |
| | | (0.044) | (0.056) | (0.055) | (0.054) | (0.051) | (0.050) |
| $\sigma_{ROA}$ | | -2.340** | -1.028 | -1.692 | -3.325** | -1.511 | -2.833** |
| | | -1.150 | -1.477 | -1.443 | -1.425 | -1.356 | -1.331 |
| $NoA^{+}$ | | -0.063* | -0.011 | -0.035 | -0.104** | -0.045 | -0.066* |
| | | (0.033) | (0.042) | (0.041) | (0.041) | (0.039) | (0.038) |
| $\sigma_{Consensus}$ | | -0.018 | -0.076 | -0.134 | -0.020 | 0.095 | -0.015 |
| | | (0.161) | (0.207) | (0.202) | (0.200) | (0.190) | (0.187) |
| $NbrBidders^{+}$ | | 0.961*** | 1.061** | 1.380*** | 0.763* | 0.721* | 0.980** |
| | | (0.362) | (0.464) | (0.454) | (0.448) | (0.426) | (0.418) |
| Diversification | | 0.028 | 0.098** | 0.045 | 0.024 | 0.011 | 0.032 |
| | | (0.032) | (0.041) | (0.040) | (0.040) | (0.038) | (0.037) |
| Tender | | -0.084 | -0.003 | 0.035 | -0.030 | -0.163* | -0.177** |
| | | (0.071) | (0.092) | (0.090) | (0.088) | (0.084) | (0.083) |
| $DealValue^{+}$ | | 0.006 | 0.014 | 0.010 | 0.005 | 0.007 | 0.004 |
| | | (0.008) | (0.011) | (0.010) | (0.010) | (0.010) | (0.010) |
| Year fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 1,469 | 1,469 | 1,469 | 1,469 | 1,469 | 1,469 | 1,469 |
| $R^2$ | 0.009 | 0.111 | 0.092 | 0.103 | 0.086 | 0.089 | 0.096 |
| Adjusted $R^2$ | 0.009 | 0.084 | 0.064 | 0.075 | 0.058 | 0.061 | 0.068 |

This table provides evidence on Hypothesis 1. It reports the regressions of bidder's tone on the method of payment and other control variables. The difference between this table and Table 4 is that we exclude hybrid deals, which reduces the sample from 5,558 to 1,469 deals. $StockforStock$ is defined in this table as a dummy variable that takes the value of one for stock-deals, and zero for cash deals. Stock deals use stocks as the only method of payments, while the other deals use only cash as the method of payment. The dependent variable for Model (1) and (2) is the sum of tone over the five semesters before the semester of the M&A announcement. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Appendix C: Multivariate Analysis of Bidders' Tone Around M&A Announcement Quarters – Inclusion of earnings management as control variable**

| Dependent Variable | Sum of Quarterly $Tone_{M\&A}$ | | Bidder's quarterly tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | 1.306*** | 1.191*** | 0.807** | 0.775* | 1.268*** | 1.158*** | 1.563*** |
| | (0.008) | (0.310) | (0.411) | (0.395) | (0.388) | (0.351) | (0.370) |
| StockforStock | 0.117*** | 0.151*** | 0.100** | 0.138*** | 0.154*** | 0.174*** | 0.137*** |
| | (0.034) | (0.035) | (0.046) | (0.044) | (0.043) | (0.039) | (0.041) |
| EarningsManagement | | 0.004** | 0.005** | 0.018** | 0.001* | 0.011** | 0.023** |
| | | (0.004) | (0.001) | (0.001) | (0.004) | (0.008) | (0.010) |
| ROA | | 0.184 | -0.380 | 0.492 | -0.222 | 0.108 | 0.358 |
| | | (0.362) | (0.480) | (0.462) | (0.454) | (0.411) | (0.432) |
| *Loss* | | -0.108*** | -0.122*** | -0.101*** | -0.125*** | -0.109*** | -0.097*** |
| | | (0.026) | (0.035) | (0.034) | (0.033) | (0.030) | (0.031) |
| Return | | -0.385*** | -0.417 | -0.344 | -0.547** | -0.121 | -0.526** |
| | | (0.196) | (0.260) | (0.251) | (0.246) | (0.223) | (0.235) |
| FE | | 0.028** | 0.013 | 0.038** | 0.023 | 0.024* | 0.028* |
| | | (0.013) | (0.017) | (0.016) | (0.016) | (0.014) | (0.015) |
| BTM | | -0.115*** | -0.089*** | -0.102*** | -0.115*** | -0.130*** | -0.111*** |
| | | (0.020) | (0.026) | (0.025) | (0.024) | (0.022) | (0.023) |
| $Size^{+}$ | | -0.007 | -0.005 | -0.008 | -0.009 | -0.007 | -0.002 |
| | | (0.006) | (0.007) | (0.007) | (0.007) | (0.006) | (0.007) |
| *Leverage* | | -0.235*** | -0.256*** | -0.267*** | -0.221*** | -0.199*** | -0.251*** |
| | | (0.040) | (0.053) | (0.051) | (0.050) | (0.045) | (0.048) |
| $\sigma_{Ret}$ | | 0.001 | -0.060*** | -0.067*** | -0.044** | 0.020 | 0.094*** |
| | | (0.018) | (0.023) | (0.022) | (0.022) | (0.020) | (0.021) |
| $\sigma_{ROA}$ | | -1.284*** | -1.448** | -0.513 | -1.302** | -1.539*** | -1.781*** |
| | | (0.475) | (0.630) | (0.606) | (0.595) | (0.539) | (0.567) |
| $NoA^{+}$ | | -0.046*** | -0.019 | -0.035* | -0.049** | -0.043** | -0.056*** |
| | | (0.016) | (0.022) | (0.021) | (0.020) | (0.019) | (0.020) |
| $\sigma_{FE}$ | | -0.074 | -0.074 | -0.106 | -0.053 | -0.029 | -0.108 |
| | | (0.083) | (0.111) | (0.106) | (0.104) | (0.095) | (0.100) |
| $NbrBidders^{+}$ | | 0.511* | 0.807** | 0.720* | 0.363 | 0.431 | 0.531 |
| | | (0.296) | (0.393) | (0.378) | (0.371) | (0.336) | (0.354) |
| Diversification | | -0.035** | -0.015 | -0.024 | -0.041** | -0.044** | -0.029 |
| | | (0.016) | (0.021) | (0.020) | (0.020) | (0.018) | (0.019) |
| Tender | | -0.117* | -0.063 | -0.054 | -0.071 | -0.177** | -0.168** |
| | | (0.062) | (0.082) | (0.079) | (0.078) | (0.070) | (0.074) |
| $DealValue^{+}$ | | -0.009** | -0.006 | -0.005 | -0.010* | -0.007 | -0.014*** |
| | | (0.004) | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| Year fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.002 | 0.089 | 0.076 | 0.074 | 0.067 | 0.076 | 0.079 |
| Adjusted $R^2$ | 0.002 | 0.081 | 0.069 | 0.067 | 0.059 | 0.069 | 0.071 |

This table provides evidence on Hypothesis 1, while controlling for earnings management. It reports the regressions of bidders' tone on the method of payment and other control variables. Stock deals use stocks as the only method of payments, while the other deals either use only cash or both cash and stock as the method of payment. The dependent variable for Model (1) and (2) is the sum of the tone over the five semesters before the semester of the M&A announcement. $StockforStock$ is a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

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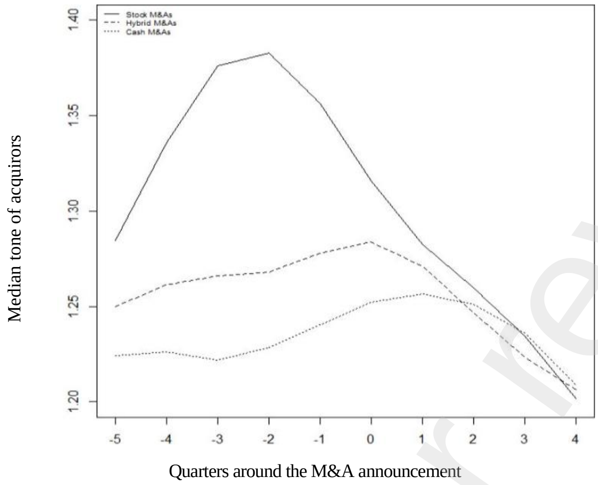
## Tables

**Figure 1.: Tone around stock-for-stock, cash and hybrid mergers and acquisitions**




This figure reports the median tone around the M&A announcement quarter (Q-5 to Q+4), were quarter zero is the quarter of the M&A announcement.

**Table 1.: Sample Distribution and Summary Statistics**

*Panel A: Distribution of sample merger and acquisition deals*

| Total | No. of Deals | Percent (%) | Nbr. of stock deals | Number of hybrid deals | Number of cash deals |
|---|---|---|---|---|---|
| 2004 | 6 | 0.108 | 4 | 1 | 1 |
| 2005 | 211 | 3.796 | 8 | 144 | 59 |
| 2006 | 267 | 4.803 | 12 | 184 | 71 |
| 2007 | 286 | 5.145 | 13 | 192 | 81 |
| 2008 | 235 | 4.228 | 17 | 162 | 56 |
| 2009 | 188 | 3.382 | 7 | 139 | 42 |
| 2010 | 323 | 5.811 | 9 | 229 | 85 |
| 2011 | 337 | 6.063 | 8 | 257 | 72 |
| 2012 | 376 | 6.765 | 11 | 282 | 83 |
| 2013 | 378 | 6.801 | 22 | 256 | 100 |
| 2014 | 466 | 8.384 | 31 | 307 | 128 |
| 2015 | 460 | 8.276 | 25 | 343 | 92 |
| 2016 | 460 | 8.276 | 26 | 368 | 66 |
| 2017 | 464 | 8.348 | 33 | 366 | 65 |
| 2018 | 459 | 8.258 | 41 | 345 | 73 |
| 2019 | 378 | 6.801 | 30 | 296 | 52 |
| 2020 | 264 | 4.749 | 5 | 217 | 42 |
| Total | 5,558 | 100.000 | 302 | 4,088 | 1,168 |

*Panel B:* Summary *statistics for the full sample*

| Variable | Mean | Standard deviation | Q1 | Median | Q3 |
|---|---|---|---|---|---|
| ROA | 0.011 | 0.025 | 0.003 | 0.011 | 0.021 |
| Loss | 0.138 | 0.345 | 0.000 | 0.000 | 0.000 |
| Return | 0.086 | 0.047 | 0.055 | 0.075 | 0.104 |
| FE | 0.019 | 0.585 | 0.002 | 0.001 | 0.034 |
| BTM | 0.443 | 0.425 | 0.229 | 0.389 | 0.620 |
| Size$^+$ | 20.783 | 79.177 | 0.789 | 2.582 | 10.153 |
| Leverage | 0.240 | 0.204 | 0.072 | 0.207 | 0.360 |
| $\sigma_{ROA}$ | 0.008 | 0.017 | 0.002 | 0.004 | 0.008 |
| $\sigma_{Ret}$ | 0.151 | 0.444 | -0.051 | 0.097 | 0.269 |
| NoA$^+$ | 9.592 | 6.382 | 6.000 | 8.264 | 11.000 |
| $\sigma_{FE}$ | 0.052 | 0.092 | 0.015 | 0.044 | 0.058 |
| NbrBidders$^+$ | 1.004 | 0.061 | 1.000 | 1.000 | 1.000 |
| Diversification | 0.458 | 0.498 | 0.000 | 0.000 | 1.000 |
| Tender | 0.015 | 0.121 | 0.000 | 0.000 | 0.000 |
| DealValue$^+$ | 1.992 | 6.382 | 0.089 | 0.255 | 0.976 |

*Panel C:* Summary *statistics for subsamples based on methods of* payment

| | *Stock deals* | | *Hybrid dear* | | *Cash deals* | |
|---|---|---|---|---|---|---|
| Variable | Mean | Median | Mean | Median | Mean | Median |
| ROA | 0.001 | 0.003 | 0.012 | 0.012 | 0.012 | 0.012 |
| Loss | 0.109 | 0.000 | 0.136 | 0.000 | 0.155 | 0.000 |
| Return | 0.080 | 0.073 | 0.087 | 0.075 | 0.087 | 0.078 |
| FE | 0.008 | 0.001 | 0.024 | 0.002 | 0.002 | 0.001 |
| BTM | 0.422 | 0.364 | 0.438 | 0.381 | 0.593 | 0.607 |
| Size$^+$ | 8.086 | 1.666 | 23.154 | 2.712 | 15.768 | 2.503 |
| Leverage | 0.155 | 0.069 | 0.246 | 0.214 | 0.242 | 0.212 |
| $\sigma_{ROA}$ | 0.006 | 0.001 | 0.008 | 0.004 | 0.008 | 0.005 |
| $\sigma_{Ret}$ | 0.101 | 0.070 | 0.149 | 0.093 | 0.169 | 0.120 |
| NoA$^+$ | 8.327 | 8.264 | 9.688 | 8.264 | 9.584 | 8.264 |
| $\sigma_{FE}$ | 0.053 | 0.032 | 0.052 | 0.047 | 0.052 | 0.036 |
| NbrBidders$^+$ | 1.010 | 1.000 | 1.001 | 1.000 | 1.010 | 1.000 |
| Diversification | 0.166 | 0.000 | 0.490 | 0.000 | 0.421 | 0.000 |
| Tender | 0.007 | 0.000 | 0.003 | 0.000 | 0.058 | 0.000 |
| DealValue$^+$ | 1.342 | 0.598 | 2.45 | 0.654 | 1.001 | 0.632 |

This table reports the distribution and summary statistics of our sample. Panel A provides the distribution of the M&A deals in our sample by year. The sample contains 5,558 deals including 302 stock deals, 4,088 hybrid deals, and 1,168 cash deals. Panel B provides the summary statistics for the deals in the full sample. Panel C splits the sample provides the summary statistics for each subsample based on the method of payment. Note that in our regression analyses, we use the logarithm of the variables indicated by a + but report the untransformed summary statistics for interpretation purposes.

**Table 2.: Correlation Table**

| Variable | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1. StockforStock | 1.000 | 0.047*** | -0.094*** | -0.020 | -0.032* | -0.005 | 0.085*** | -0.060*** |
| 2. $Tone_{M\&A}$ | 0.047*** | 1.000 | 0.083*** | -0.099*** | -0.058*** | 0.028* | -0.080*** | 0.006 |
| 3. ROA | -0.094*** | 0.083*** | 1.000 | -0.535*** | -0.201*** | -0.055*** | -0.120*** | 0.231*** |
| 4. Loss | -0.020 | -0.099*** | -0.535*** | 1.000 | 0.233*** | 0.036** | 0.029* | -0.174*** |
| 5. Return | -0.032* | -0.058*** | -0.201*** | 0.233*** | 1.000 | 0.056*** | -0.008 | -0.310*** |
| 6. FE | -0.005 | 0.028* | -0.055*** | 0.036** | 0.056*** | 1.000 | 0.003 | -0.035** |
| 7. BTM | 0.085*** | -0.080*** | -0.120*** | 0.029* | -0.008 | 0.003 | 1.000 | -0.262*** |
| 8. Size$^{+}$ | -0.060*** | 0.006 | 0.231*** | -0.174*** | -0.310*** | -0.035** | -0.262*** | 1.000 |
| 9. Leverage | -0.101*** | -0.091*** | -0.070*** | 0.102*** | 0.010 | -0.031* | -0.228*** | -0.018 |
| 10. $\sigma_{Ret}$ | -0.027* | -0.017 | 0.014 | 0.001 | 0.234*** | 0.002 | -0.118*** | 0.016 |
| 11. $\sigma_{ROA}$ | -0.032* | -0.051*** | -0.278*** | 0.260*** | 0.231*** | 0.016 | -0.104*** | -0.071*** |
| 12. NoA$^{+}$ | -0.051*** | -0.025 | 0.123*** | -0.038** | -0.066*** | -0.011 | -0.127*** | 0.591*** |
| 13. $\sigma_{FE}$ | 0.001 | -0.048*** | -0.080*** | 0.114*** | 0.088*** | 0.138*** | 0.021 | 0.041** |
| 14. NbrBidders$^{+}$ | 0.024 | 0.025 | -0.031* | 0.009 | -0.007 | -0.001 | -0.021 | 0.050*** |
| 15. Diversification | -0.141*** | -0.050*** | 0.010 | -0.028* | -0.070*** | 0.012 | 0.032* | 0.040** |
| 16. Tender | -0.016 | -0.024 | -0.033* | 0.012 | -0.023 | -0.003 | -0.019 | 0.102*** |
| 17. DealValue$^{+}$ | 0.147*** | -0.083*** | -0.097*** | 0.089*** | -0.040** | -0.012 | 0.047*** | -0.070*** |

| Variable | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|
| 1. StockforStock | -0.101*** | -0.027* | -0.032* | -0.051*** | 0.001 | 0.024 | -0.141*** | -0.016 | 0.147*** |
| 2. $Tone_{M\&A}$ | -0.091*** | -0.017 | -0.051*** | -0.025 | -0.048*** | 0.025 | -0.050*** | -0.024 | -0.083*** |
| 3. ROA | -0.070*** | 0.014 | -0.278*** | 0.123*** | -0.080*** | -0.031* | 0.010 | -0.033* | -0.097*** |
| 4. Loss | 0.102*** | 0.001 | 0.260*** | -0.038** | 0.114*** | 0.009 | -0.028* | 0.012 | 0.089*** |
| 5. Return | 0.010 | 0.234*** | 0.231*** | -0.066*** | 0.088*** | -0.007 | -0.070*** | -0.023 | -0.040** |
| 6. FE | -0.031* | 0.002 | 0.016 | -0.011 | 0.138*** | -0.001 | 0.012 | -0.003 | -0.012 |
| 7. BTM | -0.228*** | -0.118*** | -0.104*** | -0.127*** | 0.021 | -0.021 | 0.032* | -0.019 | 0.047*** |
| 8. Size$^{+}$ | -0.018 | 0.016 | -0.071*** | 0.591*** | 0.041** | 0.050*** | 0.040** | 0.102*** | -0.070*** |
| 9. Leverage | 1.000 | 0.009 | 0.039** | -0.089*** | 0.120*** | 0.001 | 0.127*** | -0.005 | 0.087*** |
| 10. $\sigma_{Ret}$ | 0.009 | 1.000 | 0.007 | 0.001 | 0.008 | -0.023 | -0.044** | -0.011 | 0.029* |
| 11. $\sigma_{ROA}$ | 0.039** | 0.007 | 1.000 | 0.002 | 0.048*** | 0.008 | -0.010 | 0.049*** | 0.016 |
| 12. NoA$^{+}$ | -0.089*** | 0.001 | 0.002 | 1.000 | 0.075*** | 0.034* | -0.074*** | 0.068*** | -0.081*** |
| 13. $\sigma_{FE}$ | 0.120*** | 0.008 | 0.048*** | 0.075*** | 1.000 | -0.010 | 0.060*** | -0.018 | 0.003 |
| 14. NbrBidders$^{+}$ | 0.001 | -0.023 | 0.008 | 0.034* | -0.010 | 1.000 | -0.027* | 0.114*** | -0.010 |
| 15. Diversification | 0.127*** | -0.044** | -0.010 | -0.074*** | 0.060*** | -0.027* | 1.000 | -0.062*** | -0.008 |
| 16. Tender | -0.005 | -0.011 | 0.049*** | 0.068*** | -0.018 | 0.114*** | -0.062*** | 1.000 | 0.061*** |
| 17. DealValue$^{+}$ | 0.087*** | 0.029* | 0.016 | -0.081*** | 0.003 | -0.010 | -0.008 | 0.061*** | 1.000 |

This table reports the Pearson (Spearman) correlation for the dependent and independent variables above (below) the diagonal. We use the logarithm of the variables indicated by a +. *, ** and *** represent the significance at the 10%, 5% and 1% levels, respectively.

**Table 3.: Average and Median Tone around M&A Announcement Quarters**

| | Stock | | Hybrid | | Cash | | Cash vs. stock | |
|---|---|---|---|---|---|---|---|---|
| Quarter | Mean | Median | Mean | Median | Mean | Median | t-test | Wilcoxon |
| Q-5 | 1.387 | 1.266 | 1.322 | 1.249 | 1.290 | 1.217 | 0.102 | 0.297 |
| Q-4 | 1.428 | 1.272 | 1.329 | 1.262 | 1.293 | 1.250 | 0.019 | 0.087 |
| Q-3 | 1.450 | 1.444 | 1.314 | 1.268 | 1.270 | 1.196 | 0.001 | 0.001 |
| Q-2 | 1.449 | 1.460 | 1.274 | 1.264 | 1.270 | 1.238 | 0.001 | 0.001 |
| Q-1 | 1.430 | 1.358 | 1.350 | 1.278 | 1.333 | 1.233 | 0.034 | 0.096 |
| Q-0 | 1.306 | 1.258 | 1.395 | 1.287 | 1.368 | 1.260 | 0.161 | 0.197 |
| Q+1 | 1.298 | 1.229 | 1.364 | 1.271 | 1.330 | 1.256 | 0.527 | 0.355 |
| Q+2 | 1.366 | 1.289 | 1.337 | 1.246 | 1.301 | 1.246 | 0.164 | 0.136 |
| Q+3 | 1.297 | 1.281 | 1.311 | 1.222 | 1.312 | 1.248 | 0.740 | 0.924 |
| Q+4 | 1.278 | 1.173 | 1.283 | 1.207 | 1.272 | 1.201 | 0.899 | 0.833 |

This table reports the tone of earnings press releases for the five quarters preceding the merger announcement, where quarter 0 is the quarter containing the merger announcement date. We also provide the average tone for four quarters after the earnings announcement quarter. The last two columns test for differences in average tone for cash and stock deals. They report the p-values for the t-statistics comparing the means and p-values for Wilcoxon signed rank tests comparing the medians.

**Table 4.: Multivariate Analysis of Bidders' Tone around M&A Announcement Quarters**

| Dependent Variable | Sum of Quarterly $Tone_{M\&A}$ | | Bidders' quarterly tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | 1.306*** | 1.190*** | 0.808** | 0.777** | 1.266*** | 1.157*** | 1.561*** |
| | (0.008) | (0.310) | (0.410) | (0.395) | (0.388) | (0.351) | (0.370) |
| StockforStock | 0.117*** | 0.152*** | 0.099** | 0.138*** | 0.155*** | 0.176*** | 0.138*** |
| | (0.034) | (0.035) | (0.046) | (0.044) | (0.043) | (0.039) | (0.041) |
| ROA | | 0.181 | -0.381 | 0.485 | -0.221 | 0.104 | 0.357 |
| | | (0.362) | (0.479) | (0.462) | (0.453) | (0.410) | (0.432) |
| Loss | | -0.108*** | -0.121*** | -0.100*** | -0.124*** | -0.109*** | -0.097*** |
| | | (0.026) | (0.035) | (0.033) | (0.033) | (0.030) | (0.031) |
| Return | | 0.370*** | 0.415 | -0.338 | 0.531** | -0.103 | 0.508** |
| | | (0.196) | (0.260) | (0.250) | (0.246) | (0.223) | (0.234) |
| FE | | 0.028** | 0.013 | 0.037$^{n}$ | 0.023 | 0.024 | 0.028* |
| | | (0.013) | (0.017) | (0.016) | (0.016) | (0.014) | (0.015) |
| BTM | | -0.115*** | -0.089*** | -0.102*** | -0.116*** | -0.130*** | -0.112*** |
| | | (0.020) | (0.026) | (0.025) | (0.024) | (0.022) | (0.023) |
| $Size^{+}$ | | -0.006 | -0.005 | -0.007 | -0.008 | -0.006 | -0.002 |
| | | (0.006) | (0.007) | (0.007) | (0.007) | (0.006) | (0.007) |
| Leverage | | -0.242*** | -0.260*** | -0.273*** | -0.229*** | -0.207*** | -0.260*** |
| | | (0.040) | (0.053) | (0.051) | (0.050) | (0.045) | (0.047) |
| $\sigma_{ROA}$ | | -1.275*** | -1.436** | -0.506 | -1.296*** | -1.529*** | -1.770*** |
| | | (0.475) | (0.629) | (0.605) | (0.595) | (0.538) | (0.567) |
| $\sigma_{Ret}$ | | 0.0003 | -0.060*** | -0.067*** | -0.044*** | 0.019 | 0.093*** |
| | | (0.018) | (0.023) | (0.022) | (0.022) | (0.020) | (0.021) |
| $NoA^{+}$ | | -0.048*** | -0.020 | -0.037* | -0.051** | -0.046** | -0.058*** |
| | | (0.016) | (0.022) | (0.021) | (0.020) | (0.018) | (0.019) |
| $\sigma_{FE}$ | | -0.074 | -0.073 | -0.105 | -0.053 | -0.029 | -0.108 |
| | | (0.083) | (0.110) | (0.106) | (0.104) | (0.095) | (0.099) |
| $NbrBidders^{+}$ | | 0.512* | 0.807** | 0.719* | 0.364 | 0.431 | 0.534 |
| | | (0.296) | (0.393) | (0.378) | (0.371) | (0.336) | (0.354) |
| Diversification | | -0.034** | -0.014 | -0.023 | -0.040** | -0.043** | -0.028 |
| | | (0.016) | (0.021) | (0.020) | (0.020) | (0.018) | (0.019) |
| Tender | | -0.117* | -0.062 | -0.053 | -0.071 | -0.177** | -0.167** |
| | | (0.062) | (0.082) | (0.079) | (0.078) | (0.070) | (0.074) |
| $DealValue^{+}$ | | -0.009** | -0.006 | -0.005 | -0.010* | -0.007 | -0.014*** |
| | | (0.004) | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| Year fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.002 | 0.089 | 0.076 | 0.075 | 0.067 | 0.076 | 0.079 |
| Adjusted $R^2$ | 0.002 | 0.082 | 0.069 | 0.067 | 0.060 | 0.069 | 0.072 |

This table provides evidence on Hypothesis 1. It reports the regressions of bidders' tone on the method of payment and other control variables. Stock deals use stocks as the only method of payment, while the other deals either use only cash or both cash and stock as the method of payment. The dependent variable for Model (1) and (2) is the sum of the tone over the five semesters before the semester of the M&A announcement. $StockforStock$ is a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 5.: Multivariate Analysis of the Bidders' Positive ($PosTone_{M\&A}$)and Negative Tone ($NegTone_{M\&A}$) around M&A Announcement Quarters**

*Panel A - Bidder's quarterly positive tone*

| Dependent Variable | Sum of Quarterly $Tone_{M\&A}$ | | Bidder's quarterly positive tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | 1.691*** | 1.736*** | 1.635*** | 1.609*** | 1.632*** | 1.434*** | 2.269*** |
| | (0.007) | (0.262) | (0.335) | (0.326) | (0.313) | (0.272) | (0.305) |
| StokforStock | 0.354*** | 0.369*** | 0.317*** | 0.348*** | 0.353*** | 0.412*** | 0.361*** |
| | (0.029) | (0.029) | (0.037) | (0.036) | (0.035) | (0.030) | (0.034) |
| Control variables | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Year fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.027 | 0.126 | 0.088 | 0.093 | 0.095 | 0.119 | 0.109 |
| Adjusted $R^2$ | 0.026 | 0.119 | 0.081 | 0.086 | 0.088 | 0.112 | 0.103 |

*Panel B - Bidder's quarterly negative tone*

| Dependent Variable | Sum of Quarterly $Tone_{M\&A}$ | | Bidder's quarterly negative tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | 0.871*** | 0.989*** | 1.088*** | 1.092*** | 1.214*** | 0.869*** | 0.783*** |
| | (0.005) | (0.157) | (0.179) | (0.179) | (0.181) | (0.185) | (0.176) |
| StockforStock | 0.007 | -0.019 | 0.007 | -0.009 | -0.030 | -0.006 | -0.030 |
| | (0.019) | (0.020) | (0.022) | (0.022) | (0.023) | (0.023) | (0.022) |
| Control variables | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Year fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.001 | 0.104 | 0.092 | 0.092 | 0.087 | 0.087 | 0.088 |
| Adjusted $R^2$ | -0.0002 | 0.096 | 0.085 | 0.085 | 0.079 | 0.079 | 0.081 |

This table reports the regressions of the bidder's positive (Panel A) and negative (Panel B) tone on the method of payment and other control variables. Stock deals use stocks as the only method of payments, while the other deals either use only cash or both cash and stock as the method of payment. The dependent variable for Model (1) and (2) is the sum of the tone over the five semesters before the semester of the M&A announcement. $StockforStock$ is a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

## Table 6.: Propensity score matching

*Panel A - Diagnostics test: Univariate test of mean differences between treatment and control groups*

| Variable | Unmatched sample | | | Matched sample | |
|---|---|---|---|---|---|
| | Mean Treatment 301 obs. | Mean Control 5,256 obs. | T-test | Mean Control 302 obs. | T-test |
| ROA | 0.001 | 0.012 | 0.000 | -0.001 | 0.528 |
| BTM | 0.593 | 0.435 | 0.000 | 0.568 | 0.517 |
| Size | 8.086 | 21.512 | 0.000 | 9.317 | 0.575 |
| $\sigma_{Ret}$ | 0.101 | 0.153 | 0.008 | 0.098 | 0.897 |
| $\sigma_{ROA}$ | 0.006 | 0.008 | 0.031 | 0.007 | 0.471 |
| Leverage | 0.155 | 0.245 | 0.000 | 0.151 | 0.814 |
| Loss | 0.109 | 0.140 | 0.101 | 0.106 | 0.896 |
| Return | 0.080 | 0.087 | 0.004 | 0.080 | 0.935 |
| FE | 0.008 | 0.019 | 0.162 | -0.005 | 0.334 |
| $\sigma_{FE}$ | 0.053 | 0.052 | 0.959 | 0.050 | 0.775 |
| *NoA* | 8.327 | 9.665 | 0.001 | 7.966 | 0.421 |

*Panel B - Tone around M&As with the matched sample*

| Quarter | *Treatment* | | *Control* | | T-test | Wilcoxon *test* |
|---|---|---|---|---|---|---|
| | Mean | Median | Mean | Median | | |
| Q-5 | 1.366 | 1.226 | 1.290 | 1.226 | 0.244 | 0.496 |
| Q-4 | 1.405 | 1.227 | 1.293 | 1.261 | 0.074 | 0.553 |
| Q-3 | 1.429 | 1.372 | 1.279 | 1.234 | 0.016 | 0.058 |
| Q-2 | 1.428 | 1.353 | 1.254 | 1.267 | 0.003 | 0.020 |
| Q-1 | 1.430 | 1.358 | 1.299 | 1.271 | 0.001 | 0.114 |
| Q-0 | 1.306 | 1.258 | 1.358 | 1.262 | 0.326 | 0.328 |

*Panel C - Regressions with the matched sample* Dependent variable

| | StockforStock | $Tone_{M\&A}$ |
|---|---|---|
| Constant | 3.515*** | 0.668 |
| | (1.206) | (0.582) |
| StockforStock | | 0.125** |
| | | (0.054) |
| ROA | 0.198*** | -0.129 |
| | (0.192) | (1.198) |
| Loss | -0.661** | -0.004 |
| | (0.260) | (0.113) |
| Return | 1.454 | -1.146 |
| | (1.679) | (0.745) |
| FE | -0.481 | -0.027 |
| | (0.399) | (0.160) |
| BTM | -0.056 | -0.201*** |
| | (0.212) | (0.063) |
| $\text{Size}^{+}$ | -0.084* | -0.020 |
| | (0.047) | (0.016) |
| Leverage | -2.874*** | -0.682*** |
| | (0.390) | (0.161) |
| $\sigma_{ROA}$ | -13.847** | -0.931 |
| | (5.748) | (1.905) |
| $\sigma_{Ret}$ | -0.195 | -0.003 |
| | (0.223) | (0.092) |
| $\text{NoA}^{+}$ | 0.074 | -0.084 |
| | (0.136) | (0.055) |
| $\sigma_{FE}$ | 0.231 | 0.063 |
| | (0.664) | (0.287) |
| $\text{NbrBidders}^{+}$ | | 1.069 |
| | | (0.709) |
| Diversification | | -0.006 |
| | | (0.060) |
| Tender | | -0.371 |
| | | (0.311) |
| $\text{DealValue}^{+}$ | | 0.003 |
| | | (0.014) |
| Year fixed effects | Yes | Yes |
| Quarter fixed effects | Yes | Yes |
| Observations | 5,558 | 604 |
| $R^2$ | 0.139 | |
| Adjusted $R^2$ | 0.080 | |
| Log Likelihood | | -931.033 |
| Akaike Inf. Crit. | | 1,940.065 |

This table provides evidence on Hypothesis 1 and reports the results using the PSM samples. Panel A reports the diagnostics test to show that there are (no) differences between the covariates of treatment and control groups before (after) the matching. Panel B compares the average of the sum of tone over the five semesters before the semester of the M&A announcement. The last two columns respectively report the p-values of the t-statistics comparing the means and the p-values of the Wilcoxon signed rank tests comparing the median. Model (1) and (2) in Panel C respectively report the regression results of the first-stage and the second-stage regression. The main independent variable of interest is $StockforStock$, which is defined as a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

# Table 7.: Entropy Balancing

*Panel A - Diagnostics test: Proof of convergence after entropy balancing*

| | Mean Treatment | Mean Control | Variance Treatment | Variance Control | Skewness Treatment | Skewness Control |
|---|---|---|---|---|---|---|
| ROA | 0.001 | 0.001 | 0.001 | 0.001 | -5.708 | -5.734 |
| Loss | 0.109 | 0.109 | 0.098 | 0.097 | 2.492 | 2.508 |
| Return | 0.080 | 0.080 | 0.001 | 0.001 | 1.798 | 1.805 |
| FE | 0.008 | 0.008 | 0.001 | 0.001 | -11.931 | -8.715 |
| BTM | 0.593 | 0.593 | 0.099 | 0.099 | 0.267 | 0.244 |
| Size | 8.086 | 8.094 | 526.829 | 534.784 | 5.339 | 5.724 |
| Leverage | 0.155 | 0.155 | 0.039 | 0.039 | 1.924 | 1.931 |
| $\sigma_{ROA}$ | 0.006 | 0.006 | 0.0003 | 0.0003 | 7.034 | 7.134 |
| $\sigma_{Ret}$ | 0.101 | 0.101 | 0.103 | 0.103 | 2.237 | 2.312 |
| NoA | 8.327 | 8.325 | 34.161 | 34.008 | 2.045 | 2.054 |
| $\sigma_{FE}$ | 0.053 | 0.053 | 0.011 | 0.011 | 7.158 | 7.199 |

*Panel B - Tone* around *M&As with the matched sample*

| | Treatment – 302 obs. | | *Control - 5,256 obs.* | | T-test | Wilcoxon test |
|---|---|---|---|---|---|---|
| Quarter | Mean | Median | Mean | Median | | |
| Q – 5 | 1.366 | 1.226 | 1.267 | 1.226 | 0.040 | 0.345 |
| Q – 4 | 1.405 | 1.227 | 1.282 | 1.226 | 0.011 | 0.015 |
| Q – 3 | 1.429 | 1.372 | 1.288 | 1.222 | 0.003 | 0.001 |
| Q – 2 | 1.428 | 1.353 | 1.257 | 1.235 | 0.002 | 0.001 |
| Q – 1 | 1.430 | 1.358 | 1.312 | 1.271 | 0.004 | 0.030 |
| Q – 0 | 1.306 | 1.258 | 1.363 | 1.264 | 0.173 | 0.200 |

*Panel C - Regressions after entropy balancing*

| Dependent variable | $Tone_{M\&A}$ |
|---|---|
| StockforStock | 0.145*** |
| | (0.034) |
| ROA | 0.811** |
| | (0.403) |
| Loss | -0.107*** |
| | (0.034) |
| Return | -0.281 |
| | (0.245) |
| FE | 0.544** |
| | (0.254) |
| BTM | -0.311*** |
| | (0.030) |
| $Size^{+}$ | -0.030*** |
| | (0.006) |
| Leverage | -0.400*** |
| | (0.043) |
| $\sigma_{ROA}$ | -0.628 |
| | (0.613) |
| $\sigma_{Ret}$ | -0.072*** |
| | (0.026) |
| $NoA^{+}$ | -0.052*** |
| | (0.017) |
| $\sigma_{FE}$ | 0.060 |
| | (0.085) |
| $NbrBidders^{+}$ | 0.774** |
| | (0.319) |
| Diversification | -0.053*** |
| | (0.016) |
| Tender | -0.032 |
| | (0.076) |
| $DealValue^{+}$ | -0.011** |
| | (0.004) |
| Year fixed effects | Yes |
| Quarter fixed effects | Yes |
| Observations | 5,558 |
| $R^2$ | 0.112 |
| Adjusted $R^2$ | 0.106 |

This table provides evidence on Hypothesis 1 and reports results using the entropy balancing method. Panel A reports the diagnostics test to show that convergence is achieved in all three dimensions (e.g. mean, variance and skewness). Panel B compares the weighted average and median of the sum of tone over the five semesters before the semester of the M\&A announcement. The last two columns respectively report the p-values of the t-statistics comparing the means and the p-values of the Wilcoxon signed rank tests comparing the median. Panel C provides the regression results of the impact of stock-for-stock deals on the sum of the bidder's tone in earnings press releases over the five quarters preceding the M&A announcement. The regression output is based on a weighted OLS, where the weights are defined by the entropy matching. The main independent variable of interest is $StockforStock$, which is defined as a dummy variable that equals one for stock deals, zero otherwise. The control variables are the same as those included in Table 4. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 8.: Instrumental Variable and Restricted Control Function Approaches – The impact of stock-for-stock M&As on tone**

| | *Panel A - Instrumental variable approach* | | *Panel B - Restricted control function* | |
|---|---|---|---|---|
| Dependent Variable | StockforStock | $Tone_{M\&A}$ | StockforStock | $Tone_{M\&A}$ |
| | (1) | (2) | (3) | (4) |
| Constant | 1.215 | 0.666* | 1.215 | 0.461 |
| | (2.106) | (0.370) | (2.106) | (0.372) |
| Inst | -0.637** | | | |
| | (0.295) | | | |
| StockforStock (fitted) | | 0.029** | | |
| | | (0.014) | | |
| StockforStock | | | | 0.231*** |
| | | | | (0.061) |
| ROA | -14.143*** | 0.404 | -14.143*** | 0.448 |
| | (3.565) | (0.371) | (3.565) | (0.372) |
| Loss | -0.425 | -0.072*** | -0.425 | -0.067** |
| | (0.277) | (0.027) | (0.277) | (0.027) |
| Return | 2.986 | -0.410** | 2.986 | -0.352* |
| | (1.947) | (0.205) | (1.947) | (0.206) |
| FE | -0.275 | 0.030** | -0.275 | 0.032** |
| | (0.425) | (0.013) | (0.425) | (0.013) |
| BTM | -0.049 | -0.132*** | -0.049 | -0.123*** |
| | (0.159) | (0.020) | (0.159) | (0.021) |
| $Size^{+}$ | 0.062 | 0.003 | 0.062 | 0.007 |
| | (0.056) | (0.006) | (0.056) | (0.006) |
| Leverage | -0.146 | -0.096** | -0.146 | -0.107** |
| | (0.538) | (0.047) | (0.538) | (0.048) |
| $\sigma_{ROA}$ | -4.186 | -0.974** | -4.186 | -0.853* |
| | (4.465) | (0.489) | (4.465) | (0.488) |
| $\sigma_{Ret}$ | -0.321 | -0.012 | -0.321 | -0.011 |
| | (0.220) | (0.018) | (0.220) | (0.018) |
| $NoA^{+}$ | -0.041 | -0.079*** | -0.041 | -0.076*** |
| | (0.159) | (0.017) | (0.159) | (0.017) |
| $\sigma_{FE}$ | 0.771 | -0.052 | 0.771 | -0.108 |
| | (0.694) | (0.089) | (0.694) | (0.090) |
| $NbrBidders^{+}$ | 3.215* | 0.650** | 3.215* | 0.624** |
| | (1.866) | (0.303) | (1.866) | (0.302) |
| Diversification | -1.411*** | -0.002 | -1.411*** | 0.005 |
| | (0.195) | (0.017) | (0.195) | (0.017) |
| Tender | -1.016 | -0.125** | -1.016 | -0.121* |
| | (0.835) | (0.063) | (0.835) | (0.063) |
| $DealValue^{+}$ | 0.257*** | -0.002 | 0.257*** | -0.006 |
| | (0.040) | (0.004) | (0.040) | (0.004) |
| GENRES | | | | -0.014*** |
| | | | | (0.005) |
| Year fixed effects | Yes | Yes | Yes | Yes |
| Quarter fixed effects | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 |
| Log Likelihood | -686.290 | | | -760.122 |
| Akaike Inf. Crit. | 1,554.579 | | | 1,704.244 |
| $R^2$ | | 0.175 | | 0.174 |
| Adjusted $R^2$ | | 0.161 | | 0.159 |
| *IV Diagnostic tests* | | | | |
| | Statistic | p-value | | |
| Weak instruments | 3.793 | 0.0515 | | |
| Wu-Hausman | 0.335 | 0.5626 | | |

This table provides evidence on Hypothesis 1 and reports the results of the instrumental variable analysis. The first step (Model 1) regresses the firm's institutional ownership percentage on the dummy variable $StockforStock$, which equals one if the deal is financed through the bidder's stock and zero otherwise. The fitted values of the first step ($StockforStock(fitted)$) are then regressed against the average tone in the second step (Model 2). In Panel B, the fitted values of ($GENRES$) in the first step are regressed against the tone of earnings press releases in the second step (Model 2). The main independent variable of interest is $StockforStock$, which is defined as a dummy variable that equals one for stock deals, zero otherwise. The control variables are the same as those included in Table 4. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 9.: Pseudo Event Study**

| Dependent Variable | Setting the event two years backward | Setting the event two years forward |
|---|---|---|
| | $Tone_{M\&A}$ | $Tone_{M\&A}$ |
| | (1) | (2) |
| Constant | 1.759*** | 1.306*** |
| | (0.386) | (0.227) |
| StockforStock | 0.071 | -0.074 |
| | (0.045) | (0.047) |
| ROA | -0.025 | 0.424 |
| | (0.356) | (0.334) |
| Loss | -0.122*** | -0.082*** |
| | (0.034) | (0.028) |
| Return | -0.408* | -0.874*** |
| | (0.222) | (0.201) |
| FE | -0.058 | -0.038 |
| | (0.049) | (0.033) |
| BTM | -0.096*** | -0.060*** |
| | (0.023) | (0.015) |
| $Size^{+}$ | -0.009 | -0.003 |
| | (0.007) | (0.006) |
| Leverage | -0.296*** | -0.297*** |
| | (0.056) | (0.048) |
| $\sigma_{ROA}$ | -0.920* | -1.296*** |
| | (0.474) | (0.382) |
| $\sigma_{Return}$ | -0.051** | -0.017 |
| | (0.026) | (0.023) |
| $NoA^{+}$ | -0.012 | -0.048** |
| | (0.021) | (0.019) |
| $\sigma_{FE}$ | -0.008 | -0.142 |
| | (0.122) | (0.091) |
| $NbrBidders^{+}$ | -0.226 | 0.143 |
| | (0.402) | (0.305) |
| Diversification | -0.033 | -0.038** |
| | (0.021) | (0.018) |
| Tender | -0.186** | -0.241*** |
| | (0.079) | (0.072) |
| $DealValue^{+}$ | -0.009* | -0.010* |
| | (0.005) | (0.006) |
| Year fixed effects | Yes | Yes |
| Quarter fixed effects | Yes | Yes |
| Observations | 3,139 | 3,961 |
| $R^2$ | 0.107 | 0.091 |
| Adjusted $R^2$ | 0.096 | 0.082 |

This table provides evidence on Hypothesis 1 and reports regressions of bidders' tone management for two sets of pseudo-events. For each stock deal and cash deal in our sample, we create a corresponding pseudo-event by shifting the announcement quarter by one year either backward (Models (1)) or forward (Models (2)). The main independent variable of interest is $StockforStock$, which is defined as a dummy variable that equals one for stock deals, zero otherwise. The control variables are the same as those included in Table 4. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 10.: The moderating effects of deal characteristics and firm monitoring on tone management around M&A deals**

| Dependent Variable | $Tone_{M\&A}$ | | | | |
|---|---|---|---|---|---|
| | *Panel A: Deal characteristic* | *Panel B: Firm performance* | | *Panel C: Firm monitoring* | |
| Factor | DealValue (log) | ROA | Return | NoA (log) | Leverage |
| | (1) | (2) | (3) | (4) | (5) |
| Constant | 1.955*** | 1.887*** | 1.848*** | 1.830*** | 1.837*** |
| | (0.338) | (0.338) | (0.338) | (0.339) | (0.338) |
| $Factor \cdot StockforStock$ | 0.065*** | -2.344** | -2.293** | -0.140** | -0.695*** |
| | (0.016) | (1.125) | (0.922) | (0.063) | (0.182) |
| StockforStock | 0.218*** | -0.010 | 0.173** | 0.280** | 0.103** |
| | (0.064) | (0.038) | (0.084) | (0.138) | (0.049) |
| ROA | 0.618 | 0.854** | 0.582 | 0.637 | 0.696* |
| | (0.395) | (0.410) | (0.395) | (0.395) | (0.395) |
| Loss | -0.119*** | -0.117*** | -0.120*** | -0.119*** | -0.118*** |
| | (0.029) | (0.029) | (0.029) | (0.029) | (0.029) |
| Return | -0.550** | -0.543** | -0.483** | -0.544** | -0.555*** |
| | (0.214) | (0.214) | (0.216) | (0.214) | (0.214) |
| FE | 0.013 | 0.014 | 0.012 | 0.013 | 0.013 |
| | (0.014) | (0.014) | (0.014) | (0.014) | (0.014) |
| BTM | -0.093*** | -0.094*** | -0.095*** | -0.094*** | -0.092*** |
| | (0.021) | (0.021) | (0.021) | (0.021) | (0.021) |
| $Size^{+}$ | -0.008 | -0.009 | -0.008 | -0.009 | -0.007 |
| | (0.006) | (0.006) | (0.006) | (0.006) | (0.006) |
| Leverage | -0.238*** | -0.248*** | -0.253*** | -0.248*** | -0.216*** |
| | (0.043) | (0.043) | (0.043) | (0.043) | (0.044) |
| $\sigma_{Ret}$ | 0.104*** | 0.103*** | 0.104*** | 0.103*** | 0.102*** |
| | (0.019) | (0.019) | (0.019) | (0.019) | (0.019) |
| $\sigma_{ROA}$ | -0.996* | -1.139** | -1.026** | -1.039** | -0.990* |
| | (0.518) | (0.520) | (0.518) | (0.518) | (0.518) |
| $NoA^{+}$ | -0.061*** | -0.061*** | -0.062*** | -0.055*** | -0.062*** |
| | (0.018) | (0.018) | (0.018) | (0.018) | (0.018) |
| $\sigma_{FE}$ | -0.097 | -0.111 | -0.088 | -0.091 | -0.095 |
| | (0.091) | (0.091) | (0.091) | (0.091) | (0.091) |
| $NbrBidders^{+}$ | 0.303 | 0.272 | 0.250 | 0.283 | 0.273 |
| | (0.323) | (0.324) | (0.323) | (0.324) | (0.323) |
| Diversification | -0.016 | -0.018 | -0.014 | -0.016 | -0.016 |
| | (0.017) | (0.017) | (0.017) | (0.017) | (0.017) |
| Tender | -0.084 | -0.087 | -0.090 | -0.089 | -0.090 |
| | (0.068) | (0.068) | (0.068) | (0.068) | (0.068) |
| $DealValue^{+}$ | -0.026*** | -0.022*** | -0.022*** | -0.022*** | -0.023*** |
| | (0.005) | (0.005) | (0.005) | (0.005) | (0.005) |
| Year fixed effects | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | Yes | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.082 | 0.080 | 0.081 | 0.080 | 0.082 |
| Adjusted $R^2$ | 0.075 | 0.073 | 0.073 | 0.073 | 0.075 |

This table provides evidence on Hypothesis 2 (Panel A), Hypothesis 3 (Panel B) and Hypothesis 4 (Panel C). In Panel A, Model (1) focuses on the impact of the deal's value ($DealValue$), which is obtained from the SDC database. In Panel B, Model (2) and (3) focus on the impact of firm performance, which is respectively measured based on the firm's return on assets ($ROA$) and past stock returns ($Return$). In Panel C, Model (4) and (5) respectively proxy firm monitoring with analysts' coverage and the proportion of leverage in the firm. The main independent variable of interest $StockforStock$ is, which is defined as a dummy variable that equals one for stock deals, zero otherwise. The control variables are the same as those included in in Table 4. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 11.: Impact of tone on the bidder's stock return around the earnings announcement**

| | *Cumulative abnormal returns* | |
|---|---|---|
| Dependent Variable | CAR[-1,+1] | CAR[-3,+3] |
| | (1) | (2) |
| Constant | 0.006 | 0.006 |
| | (0.018) | (0.021) |
| $AbTone_{M\&A}$ | 0.003*** | 0.004*** |
| | (0.0005) | (0.001) |
| $ExpectedTone_{M\&As}$ | -0.002 | 0.001 |
| | (0.002) | (0.002) |
| ROA | 0.012 | 0.019 |
| | (0.010) | (0.012) |
| Loss | -0.003*** | -0.003*** |
| | (0.001) | (0.001) |
| FE | 0.156*** | 0.176*** |
| | (0.016) | (0.018) |
| BTM | 0.002 | 0.001 |
| | (0.003) | (0.003) |
| $\sigma_{Ret}$ | -0.002*** | -0.004*** |
| | (0.001) | (0.001) |
| $\sigma_{ROA}$ | 0.0002 | 0.004 |
| | (0.013) | (0.015) |
| $NoA^{+}$ | -0.001* | -0.001** |
| | (0.0005) | (0.001) |
| $\sigma_{FE}$ | -0.003 | -0.006* |
| | (0.003) | (0.004) |
| Year fixed effects | Yes | Yes |
| Quarter fixed effects | Yes | Yes |
| Observations | 27,790 | 27,790 |
| $R^2$ | 0.010 | 0.011 |
| Adjusted $R^2$ | 0.009 | 0.009 |

This table reports the impact of tone management on bidder's premerger stock returns. The dependent variable is the bidder's cumulative abnormal stock returns around the earnings announcement day. The dependent variable of Model (1) is the cumulative abnormal return from one day before to one day after the earnings announcement day ($CAR[-1,+1]$). Model (2)'s dependent variable is the cumulative abnormal return from three days before to three days after the earnings announcement day ($CAR[-3,+3]$).The independent variables include earnings characteristics as well as firm characteristics. $AbTone$ is the difference between the actual tone and the expected tone ($ExpectedTone_{M\&As}$) estimated from Equation 7. Year fixed effects and industry fixed effects are included. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 12.: Stock-for-Stock merger and acquisition announcement, tone and bidder's long-term stock returns after the merger**

| | *Cumulative abnormal returns* | | | |
|---|---|---|---|---|
| Dependent Variable | CAR[-3,+3] (1) | CAR[+1,+60] (2) | CAR[+1,+180] (3) | CAR [+1, +360] (4) |
| Constant | 0.029 | -0.0002 | 0.007 | -0.182 |
| | (0.030) | (0.056) | (0.086) | (0.142) |
| $StockforStock \cdot AbTone_{M\&A}$ | 0.015** | 0.012 | -0.031* | -0.052** |
| | (0.006) | (0.011) | (0.018) | (0.022) |
| StockforStock | -0.023*** | -0.029*** | -0.040*** | -0.048** |
| | (0.005) | (0.008) | (0.013) | (0.022) |
| AbTone | 0.004** | 0.013*** | 0.030*** | 0.030*** |
| | (0.002) | (0.003) | (0.005) | (0.008) |
| ROA | -0.050 | 0.239*** | 0.505*** | 0.533** |
| | (0.047) | (0.087) | (0.134) | (0.222) |
| Loss | -0.004 | -0.014** | -0.021** | -0.019 |
| | (0.003) | (0.006) | (0.010) | (0.016) |
| Return | 0.041 | -0.033 | 0.122* | 0.310** |
| | (0.026) | (0.048) | (0.073) | (0.122) |
| FE | 0.0004 | -0.005* | -0.007 | -0.022*** |
| | (0.002) | (0.003) | (0.005) | (0.008) |
| BTM | -0.011*** | -0.012** | -0.006 | 0.015 |
| | (0.003) | (0.005) | (0.007) | (0.012) |
| $Size^+$ | -0.003*** | -0.003** | -0.002 | -0.006 |
| | (0.001) | (0.001) | (0.002) | (0.003) |
| Leverage | -0.013** | -0.006 | -0.010 | -0.011 |
| | (0.005) | (0.010) | (0.015) | (0.025) |
| $\sigma_{Ret}$ | 0.032*** | 0.063*** | 0.032*** | 0.095*** |
| | (0.002) | (0.004) | (0.007) | (0.014) |
| $\sigma_{ROA}$ | 0.085 | -0.031 | -0.003 | 0.387 |
| | (0.062) | (0.114) | (0.176) | (0.292) |
| $NoA^+$ | 0.002 | 0.00003 | 0.001 | 0.010 |
| | (0.002) | (0.004) | (0.006) | (0.010) |
| $\sigma_{FE}$ | -0.004 | -0.020 | -0.027 | -0.115** |
| | (0.011) | (0.020) | (0.031) | (0.051) |
| $NbrBidders^+$ | 0.354 | 0.172 | 0.253 | 0.182 |
| | (0.323) | (0.326) | (0.423) | (0.222) |
| Diversification | -0.002 | -0.005 | 0.001 | 0.005 |
| | (0.002) | (0.004) | (0.006) | (0.010) |
| Tender | -0.010 | -0.014 | -0.015 | 0.001 |
| | (0.008) | (0.015) | (0.023) | (0.039) |
| $DealValue^+$ | 0.003*** | 0.001 | -0.001 | -0.002 |
| | (0.001) | (0.001) | (0.002) | (0.003) |
| Year fixed effects | Yes | Yes | Yes | Yes |
| Quarter fixed effects | Yes | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.069 | 0.071 | 0.047 | 0.055 |
| Adjusted $R^2$ | 0.061 | 0.063 | 0.039 | 0.047 |

This table analyzes the bidder's stock returns around the M&A announcement as well as the stock returns in the post-merger period. The dependent variable in this table is the bidder's cumulative abnormal stock returns around the M&A announcement day. In Model (1), the dependent variable around the M&A announcement is the cumulative abnormal return three days before to three days after the M\&A announcement day. In Model (2), the dependent variable focuses on the cumulative abnormal returns over sixty days after the M&A announcement day, starting one day after the M&A announcement day. Model (3) focuses on the cumulative abnormal returns for a 180 day period, starting one day after the M&A announcement; Model (4) has as dependent variable the cumulative abnormal return over 360 days, starting one day after the M&A announcement day. The main variable of interest in this table is the interaction between $StockforStock$ and $AbTone$. $StockforStock$ is defined as a dummy variable that equals one for stock deals, zero otherwise. $AbTone$ is the difference between the actual tone and the expected tone ($ExpectedTone_{M\&As}$) estimated from Equation 7. The independent variables include earnings and deal characteristics, as well as firm characteristics. Year fixed effects and industry fixed effects are included. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 13.: Multivariate Analysis of Targets' Tone around M&A Announcement Quarters**

| Dependent Variable | Sum of Quarterly $Tone_{M\&A}$ | | Targets' quarterly tone around M&A announcement | | | | |
|---|---|---|---|---|---|---|---|
| | | | Quarter -5 | Quarter -4 | Quarter -3 | Quarter -2 | Quarter -1 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Constant | -0.001* | 0.013 | 14.311 | 12.463 | 12.767 | 4.331 | 10.798 |
| | (0.001) | (0.020) | (9.186) | (3.503) | (3.121) | (3.397) | (4.968) |
| $StockforStock$ | -0.002 | -0.023 | 7.677 | 0.703 | 9.150 | 1.045 | 2.590 |
| | (0.005) | (0.020) | (9.388) | (3.579) | (3.190) | (3.472) | (5.077) |
| $ROA$ | | 0.037 | 17.848 | -6.466 | -17.739 | -2.073 | 53.570 |
| | | (0.082) | (38.275) | (14.594) | (13.005) | (14.154) | (20.699) |
| $Loss$ | | 0.005 | 1.380 | -0.832 | -0.584 | -0.144 | 2.279 |
| | | (0.004) | (2.007) | (0.765) | (0.682) | (0.742) | (1.085) |
| $Return$ | | 0.007 | 2.624 | -9.041 | 0.707 | -3.137 | 10.635 |
| | | (0.034) | (15.724) | (5.995) | (5.343) | (5.815) | (8.503) |
| $FE$ | | -0.0001 | 0.048 | -0.025 | 0.010 | 0.006 | 0.098 |
| | | (0.0002) | (0.087) | (0.033) | (0.030) | (0.032) | (0.047) |
| $BTM$ | | 0.00004 | -0.003 | 0.006 | -0.007 | 0.008 | -0.007 |
| | | (0.00003) | (0.016) | (0.006) | (0.005) | (0.006) | (0.009) |
| $Size^{+}$ | | -0.001 | 0.063 | -0.623 | -0.098 | -0.181 | -0.004 |
| | | (0.001) | (0.389) | (0.148) | (0.132) | (0.144) | (0.211) |
| $Leverage$ | | 0.015 | -9.486 | -1.255 | -7.173 | -0.281 | -7.514 |
| | | (0.019) | (8.945) | (3.410) | (3.039) | (3.308) | (4.837) |
| $\sigma_{Return}$ | | -0.010 | 2.122 | 1.395 | 1.851 | 1.001 | 1.698 |
| | | (0.003) | (1.250) | (0.477) | (0.425) | (0.462) | (0.676) |
| $\sigma_{ROA}$ | | -0.041 | -82.096 | -45.948 | -78.366 | -23.436 | -134.559 |
| | | (0.228) | (106.613) | (40.650) | (36.224) | (39.424) | (57.655) |
| $NoA^{+}$ | | 0.002 | -1.903 | 0.034 | -1.491 | -0.258 | -2.177 |
| | | (0.005) | (2.256) | (0.860) | (0.766) | (0.834) | (1.220) ) |
| $\sigma_{FE}$ | | -0.041 | 14.645 | -8.439 | 3.850 | -0.122 | 27.624 |
| | | (0.056) | (26.180) | (9.982) | (8.895) | (9.681) | (14.158) |
| $NbrBidders^{+}$ | | -0.016 | -7.598 | -5.588 | -5.582 | -1.154 | -2.380 |
| | | (0.012) | (5.612) | (2.140) | (1.907) | (2.075) | (3.035) |
| $Diversification$ | | 0.0004 | 0.236 | -0.306 | -0.260 | 0.876 | 0.503 |
| | | (0.002) | (0.774) | (0.295) | (0.263) | (0.286) | (0.419) |
| $Tender$ | | 0.008 | -2.326 | -0.460 | -2.080 | 0.128 | -2.040 |
| | | (0.005) | (2.295) | (0.875) | (0.780) | (0.849) | (1.241) |
| $DealValue^{+}$ | | -0.0005 | 0.420 | -0.057 | 0.346 | -0.080 | 0.328 |
| | | (0.001) | (0.491) | (0.187) | (0.167) | (0.181) | (0.265) |
| Year fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Quarter fixed effects | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 37 | 37 | 37 | 37 | 37 | 37 | 37 |
| $R^2$ | 0.006 | 0.996 | 0.963 | 0.994 | 0.996 | 0.994 | 0.986 |
| Adjusted $R^2$ | -0.023 | 0.857 | -0.346 | 0.794 | 0.848 | 0.778 | 0.48 |

This table reports the regressions of the target's tone on the method of payment and other control variables. Stock deals use stocks as the only method of payments, while the other deals either use only cash or both cash and stock as the method of payment. The dependent variable for Model (1) and (2) is the sum of the tone over the five semesters before the semester of the M&A announcement. $StockforStock$ is a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.

**Table 14.: Multivariate Analysis of the Bidder's Tone Dispersion ($ARF$) for Net ($Tone_{M\&A}$), Positive ($PosTone_{M\&A}$) and Negative Tone ($NegTone_{M\&A}$) before M&A Announcement Quarters**

| *Dependent variable* | *ARF* | | |
|---|---|---|---|
| | $Tone_{M\&A}$ (1) | $PosTone_{M\&A}$ (2) | $NegTone_{M\&A}$ (3) |
| Constant | 0.530*** | 0.511*** | 0.548*** |
| | (0.014) | (0.014) | (0.029) |
| *StockforStock* | 0.002* | 0.003* | 0.001 |
| | (0.001) | (0.001) | (0.003) |
| *ROA* | 0.009* | 0.018* | 0.047 |
| | (0.004) | (0.008) | (0.029) |
| *Loss* | -0.001 | -0.001 | 0.002 |
| | (0.001) | (0.001) | (0.002) |
| *Return* | -0.003 | 0.008 | 0.016 |
| | (0.008) | (0.008) | (0.016) |
| *FE* | 0.004 | -0.003 | -0.004 |
| | (0.005) | (0.005) | (0.001) |
| *BTM* | 0.001 | -0.005 | -0.005*** |
| | (0.001) | (0.001) | (0.002) |
| $Size^{+}$ | -0.004* | -0.003 | -0.002*** |
| | (0.002) | (0.002) | (0.005) |
| *Leverage* | -0.001 | -0.003 | -0.006* |
| | (0.002) | (0.002) | (0.004) |
| $\sigma_{Return}$ | 0.001* | -0.001 | 0.001 |
| | (0.001) | (0.001) | (0.001) |
| $\sigma_{ROA}$ | -0.032* | 0.019 | -0.041 |
| | (0.019) | (0.018) | (0.038) |
| $NoA^{+}$ | 0.004 | -0.001 | 0.002 |
| | (0.001) | (0.001) | (0.001) |
| $\sigma_{Consensus}$ | -0.002 | 0.003 | -0.002 |
| | (0.003) | (0.003) | (0.007) |
| $NbrBidders^{+}$ | 0.002 | 0.001 | -0.010 |
| | (0.012) | (0.011) | (0.024) |
| *Diversification* | -0.004 | -0.001 | 0.001 |
| | (0.001) | (0.001) | (0.001) |
| *Tender* | -0.001 | 0.0004 | 0.005 |
| | (0.002) | (0.002) | (0.005) |
| $DealValue^{+}$ | -0.001 | -0.003* | -0.0002 |
| | (0.002) | (0.002) | (0.003) |
| Year fixed effects | Yes | Yes | Yes |
| Quarter fixed effects | Yes | Yes | Yes |
| Observations | 5,558 | 5,558 | 5,558 |
| $R^2$ | 0.027 | 0.026 | 0.046 |
| Adjusted $R^2$ | 0.010 | 0.008 | 0.029 |

This table reports the regressions of the bidder's tone dispersion on the method of payment and other control variables. Stock deals use stocks as the only method of payments, while the other deals either use only cash or both cash and stock as the method of payment. The dependent variable for Model (1) and (2) is the average reduced frequency ($ARF$) of net ($Tone_{M\&A}$), positive ($PosTone_{M\&A}$) and negative tone ($NegTone_{M\&A}$) over the five semesters before the semester of the M&A announcement. $StockforStock$ is a dummy variable that equals one for stock deals, zero otherwise. All continuous variables are winsorized at the 1% and 99% percentiles. The significance of the coefficients is tested using Newey-West standard errors, which are reported in parentheses. *, ** and *** represent the significance at the 10%, 5% and 1% confidence levels, respectively, based on a two-sided *t*-test. Goodness of fit is evaluated by $R^2$ and adj. $R^2$.